

TOPICAL REVIEW

A Comprehensive Review of Recent Advances in Multimodal Multimedia Indexing and Retrieval

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ABSTRACT With the exponential growth of digital data across several domains, multimedia retrieval has emerged as a very critical research area. While much development has been witnessed in feature extraction, indexing methods, and deep learning-based retrieval models, many challenges persistently hamper the creation of efficient and scalable multimedia retrieval systems. This review systematically analyzes recent research in multimedia retrieval, highlighting key methodologies, findings, and limitations across various studies. The major gaps identified include the semantic disparity between low-level features and high-level semantics, as well as challenges of large datasets, privacy and security concerns, and the explainability of deep learning models. Besides, challenges on noisy and imbalanced data, multimodal data sources, and a lack of standardized benchmarking frameworks further limit the performance of existing systems. This paper comprehensively presents these gaps and proposes future research directions to bridge them, thereby directing the development of more robust, scalable, and user-centered multimedia retrieval systems. To overcome persistent challenges in versatility, semantic alignment, and trustworthiness, we introduce a unified framework that seamlessly integrates multimodal semantic fusion, explainable AI, privacy-preserving learning, and continual adaptation for robust and future-ready multimedia retrieval.

INDEX TERMS Multimedia indexing, multimedia retrieval, semantic gap, real-time retrieval, user interaction, cross-domain applicability, noisy data.

I. INTRODUCTION

Advancements in multimedia indexing and retrieval systems are necessary to manage and exploit the vast volume of information generated by the exponential growth of digital content. This poses a great challenge for handling diverse data types, bridging the semantic gaps, and developing effective indexing structures. However, these developments have left some knowledge gaps that hinder the development of robust search systems.

Digital content management depends significantly on multimedia indexing and retrieval mechanisms. Content-Based Image Retrieval (CBIR) emerged early on as an alternative retrieval method that used image features instead of conventional metadata systems [1]. The Q.B.I.C system functioned as one of the earliest CBIR approaches through its

visual content indexing methods, which exposed significant limitations in low-level visual indexing.

Since then, the field has experienced substantial change. Users now require more powerful systems to handle multiple media types and support elaborate query requests. The fundamental challenge in multimedia retrieval arises when low-level features, such as color or texture, fail to map to high-level user-relevant concepts like objects or scenes. This semantic gap continues to hinder the accurate retrieval of multimedia. Smeulders et al. [2] highlighted this early, and Datta et al. [3] reaffirmed its significance in more recent evaluations of image retrieval.

A deficiency still exists in current strategies that focus narrowly on individual types of multimedia. These systems typically fail to unify operations across text, image, audio, and video data. Lew et al. [4] emphasized this in their survey on multimedia information retrieval, advocating for systems that can handle diverse modalities. Snoek and Worring [5]

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further reinforced that multimodal integration is essential for enhancing video retrieval performance.

At the same time, indexing structures must be scalable to manage large and growing databases. While Flickner et al. [1] showcased the benefits of early indexing methods, their limitations have become more evident with increased data volume. Naphade et al. [6] introduced a large-scale multimedia ontology to support semantic retrieval, but flexibility and scalability challenges persist.

User interaction and cross-domain applicability remain underexplored. More intuitive and interactive systems are needed, as highlighted by Jaimes and Sebe [7], who identified the importance of human-centered multimedia retrieval. Rasiwasia et al. [14] presented a cross-modal retrieval system to address domain variability, yet real-world inconsistencies and noisy data still require robust processing techniques. A comprehensive survey of constrained multiobjective evolutionary algorithms (CMOEAs) is presented in [8], categorizing existing methods, benchmarks, and constraint-handling techniques. It also highlights real-world applications, current challenges, and future research directions in evolutionary constrained multiobjective optimization.

This paper examines recent advancements in multimedia indexing and retrieval. It contends that future research must address the above gaps to move toward complete and integrated retrieval systems. Many current systems perform well for individual media types but struggle to handle hybrid systems that combine text, image, audio, and video.

Other critical areas include bridging the semantic gap, building scalable and efficient indexing structures, enabling real-time retrieval, and supporting intuitive user interactions. Cross-domain robustness and handling noisy data are also key open problems. Although significant progress has been made, further creative and integrated research is essential to build comprehensive and deployable retrieval systems.

Traditional indexing techniques relied on handcrafted low-level features, which failed to capture the rich semantics of visual and textual content. Deep learning-based retrieval systems have since enhanced representational capacity, allowing for better alignment across modalities. Still, issues like scalability, semantic mismatch, and lack of explainability continue to affect system reliability.

Recent advancements in foundation models—such as CLIP, BLIP-2, and GPT-4—and the rise of prompt engineering have transformed the retrieval landscape. These models demonstrate strong cross-modal understanding through instruction-tuned architectures. At the same time, emerging paradigms such as edge computing and IoT demand lightweight, real-time, and privacy-preserving retrieval solutions. These developments challenge traditional indexing principles and call for a complete rethinking of how scalable, explainable, and semantically aware retrieval systems should be built.

Problem Statement: Despite decades of progress in multimedia retrieval, existing systems still fall short in meeting

the evolving demands of real-world applications. Traditional models often specialize in isolated modalities and rely heavily on handcrafted features, failing to bridge the semantic gap between low-level representations and high-level user intent. More recent multimodal systems—while powerful—are often siloed by domain, lack generalizability, and struggle to balance explainability, computational cost, and retrieval performance. Furthermore, benchmark fragmentation and inconsistent evaluation protocols hinder reproducibility and comparative assessment across models. There is a lack of a consolidated, technically grounded, and forward-looking review that maps these limitations to concrete architectural and empirical gaps.

This paper addresses this void by systematically analyzing recent advances in multimodal retrieval, evaluating model-specific innovations, and highlighting open challenges that obstruct scalable, interpretable, and semantically robust multimedia indexing and retrieval frameworks.

A. CONTRIBUTIONS AND SCOPE

This paper makes the following key contributions:

- It presents a comparative analysis and critical synthesis (Section III) of state-of-the-art multimodal retrieval systems, highlighting trade-offs between performance, scalability, explainability, and modality support.
- It introduces a novel conceptual taxonomy of multimodal retrieval architectures (Section IV), classifying models based on fusion strategy, modality coverage, and semantic depth to provide a clearer understanding of architectural evolution.
- It compiles detailed comparative tables that analyze model-specific innovations, including fusion mechanisms, parameter configurations, and benchmark accuracies.
- It identifies persistent research gaps—such as semantic misalignment, benchmark fragmentation, and cross-domain generalization—and proposes future directions aligned with real-world deployment scenarios.

This review is limited to academic, open-access models and datasets; commercial, proprietary systems and closed-source evaluations are excluded. Experimental validation is not conducted, as this work is intended as a critical synthesis and roadmap rather than a benchmark contribution. Instead, the emphasis is on thematic analysis, architectural taxonomy, and research guidance.

The remainder of the paper is structured as follows: Section II reviews the existing literature on multimedia indexing and retrieval. Section III presents a comparative analysis and critical synthesis. Section IV proposes a novel conceptual taxonomy of multimodal retrieval architectures. Section V highlights key research gaps in current systems. Section VI examines the trade-off between computational efficiency and retrieval accuracy. Section VII discusses Benchmark Datasets and Evaluation Metrics. Section VIII provides a critical discussion of model capabilities, highlights domain-specific

trade-offs, and outlines open research questions. Section IX concludes the paper with a summary of key insights, and Section X proposes directions for future research.

II. LITERATURE REVIEW

Multimedia indexing and retrieval are fundamental components necessary for the efficient and optimal management of the vast and diverse multimedia content that characterizes contemporary digital environments. The diversity of multimedia data, encompassing images, videos, and audio, presents significant challenges and opportunities in domains such as entertainment, education, healthcare, and digital libraries. Effective multimedia retrieval systems are essential for straightforward access to relevant information and for supporting advanced applications such as content-based recommender systems and real-time multimedia analysis in industries ranging from media production to medical diagnostics.

In the late 20th century, multimedia retrieval was primarily based on basic keyword-driven systems that indexed and retrieved content using text-based metadata. With advances in computational techniques, these systems evolved to incorporate more sophisticated methods, including color histograms and audio spectrograms, which paved the way for content-based image retrieval (CBIR). The development of standardized reference databases significantly accelerated research in this area by providing benchmarks to evaluate and enhance retrieval algorithms.

Metadata, which may include information such as author details, creation or modification timestamps, and descriptive comments, plays a crucial role in the indexing of multimedia content. Modern approaches to large-scale data processing focus on feature extraction, a process that transforms raw multimedia data into structured, high-level representations that facilitate efficient retrieval. Advanced techniques enable users to articulate their search requirements more precisely, which in turn helps computational systems interpret these queries more accurately.

Relevance ranking techniques, which involve computing relevance scores and ranking multimedia content based on semantic similarity, are central to optimizing the retrieval process. These techniques enhance the ability to search through large multimedia collections, ensuring that the most pertinent results are presented to the user on time. To meet the demands of modern multimedia search, it is essential to build upon these foundational concepts by enabling large-scale searches, providing time-efficient solutions for real-time multimedia retrieval, and seamlessly integrating various types of multimedia data.

These advancements enhance the user experience by enabling the rapid and accurate retrieval of multimedia content, thereby expanding the potential applications of digital multimedia in various fields. This section presents a comprehensive review of recent research contributions, organized into thematic categories for clarity and coherence.

A. CONTENT-BASED IMAGE RETRIEVAL (CBIR)

CBIR systems attempt to boost retrieval accuracy by linking visual features with user search intent. Early systems placed strong emphasis on manual feature extraction, but current research centers on deep learning platforms and cross-technique combinations to enhance all aspects of feature discovery and information retrieval operations. The system has progressed, but ongoing obstacles, including semantic processing capabilities, large-scale implementation, and computing speed, persist. Table 1 summarizes research papers based on CBIR.

CBIR technology evolved because of more advanced content-based feature acquisition and machine learning strategies. Datta et al. [3] examined early CBIR techniques in their review to showcase development while explaining remaining semantic gap problems. Ashraf et al. [9] achieved higher accuracy through low-level feature combination, and Srivastava et al. [10] focused on deep learning methods despite computational complexities. Pradhan et al. [11] utilized DNA-based encoding to decrease feature dimensions, while Sukhia et al. [12] developed rotation-invariant descriptors for achieving robust retrieval. This research improves the performance of CBIR systems while making it clear that semantic deficiencies and efficiency limitations remain significant challenges.

B. MULTIMODAL RETRIEVAL SYSTEMS

Modern retrieval systems handle the complex integration of three or more data types, including text, images, audio, and video, to create complete information searches. Research efforts have strived to boost retrieval accuracy and robustness by developing frameworks that unite multiple data modalities. The existing improvements have been clear; however, data alignment difficulties, assessment metric standardization, and system-efficient real-world deployment issues still need resolution. Table 2 summarizes research papers based on modern retrieval systems.

The development of Multimedia Information Retrieval (MIR) occurred through techniques that unite multiple media formats while solving semantic problems. Michael et al. [13] and Snoek et al. [5] employed content-based and multimodal retrieval methods to enhance accuracy rates while facing constraints related to scalability and data alignment. Rasiwasia et al. [14] applied CCA to create cross-modal correlation approaches for text-to-image retrieval, resulting in significant performance improvements. Despite their computational requirements, Mahmood et al. [15] developed a neural network system that increased precision and recall rates. Spoken content retrieval was the focus of Bendib and Laouar research [16], where they used partial transcription and semantic enrichment to highlight the significance of linguistic resources. These studies showcase multimodal MIR systems with semantic enhancements, yet acknowledge that applicability to large, dynamic datasets and integration remain active research areas.

C. HASHING AND INDEXING TECHNIQUES

Since hashing and indexing techniques are used in large-scale media datasets, they are crucial to the speed and effectiveness of multimedia database retrieval. To improve retrieval accuracy while lowering computational costs, current research efforts focus on improving hash code generation, optimizing indexing structures, and integrating machine learning. Existing problems with system stability and scale, as well as control parameter challenges, have not been addressed by the most recent indexing solutions and hash code enhancements. Table 3 summarizes research papers based on hashing and indexing techniques.

Research in image retrieval has emphasized the development of efficient hashing methods because they improve both extensibility and accuracy levels. Fourier-based binary hashing produced high accuracy in the work conducted by Ahmad et al. [17], though their method faced difficulties when generating short codes. Wang et al. [18] proposed Deep Index with probabilistic quantization, which generated excellent outcomes but faced extensibility problems. UDBH was developed by Guo et al. [19] with a ViT backbone structure, delivering high computational accuracy. Jin et al. [20] created RCSH, which performed well in generalization but struggled with unbalanced datasets. Jiang et al. [21] proposed ROH for cross-modal retrieval, which was effective with long codes but sensitive to parameters.

D. PRIVACY-PRESERVING RETRIEVAL SYSTEMS

The objective of retrieval systems built with privacy protection is to combine security features with user privacy safeguards without compromising retrieval effectiveness.

Different multimedia retrieval frameworks now use three primary techniques, consisting of encryption-based retrieval and hashing systems, together with differential privacy methods. Security enhancements through these methods remain a barrier to their universal implementation because of problems with scalability, together with adaptability issues and complex computational requirements.

Table 4 summarizes research papers based on the privacy-preserving context. Research in privacy-preserving image retrieval now demonstrates both effective secure retrieval algorithms as well as optimized method implementations. The research by Zhang et al. demonstrated accurate image retrieval through eLSH combined with secret sharing, but its expandability features require further improvement [22]. Hyperchaotic systems combined with AES encryption provide strong security together with high mAP according to Anyu et al. [23], although the computational cost remains high. The researchers at Ma et al. [24] developed TPE for IoT applications while maintaining accurate results without significant file expansion, but extensibility problems continue to emerge.

The implemented techniques improve security levels yet require additional advancement work to serve large-scale execution requirements. In this paper, we distinguish

privacy-preserving retrieval as approaches that protect user query or content visibility from external entities (e.g., via encryption or anonymization), while *secure retrieval* refers to broader mechanisms that ensure data integrity, access control, and protection from unauthorized access.

E. MACHINE LEARNING AND DEEP LEARNING-BASED APPROACHES

The implementation of machine learning combined with deep learning techniques transformed multimedia retrieval by developing better feature extraction techniques and representation methods, and indexing procedures. Advanced computer science technology enabled researchers to develop transformer architecture and convolutional networks (CNNs) as neural network models, which enhanced retrieval accuracy. Additional work is needed to improve the technology because it faces challenges regarding interpretability as well as scalability, and computational resource issues. Table 5 summarizes research papers based on Machine Learning and Deep Learning-Based Approaches. Neural, along with transformer models, are central to present-day IR developments because they enhance both retrieval speed and precision according to recent advancements. The researchers of Liu et al [25] enhanced policy evaluation through the integration of decision trees and ABAC. According to Mahmood et al. [15], neural networks enhance multimedia retrieval, although they encounter scalability problems. Rateria and Singh [26] used BERT and WMD for accurate, low-resource retrieval. Modern IR models have led to improved precision as researched by Hambarde and Proenca [27], yet these models consume high amounts of data and computing resources. Chen et al.'s [28] work delivered the best image retrieval results by combining CNNs with transformers.

F. SCALABILITY AND EFFICIENCY ENHANCEMENTS

Large-scale multimedia retrieval systems face important obstacles regarding both efficiency and malleability problems. Research studies now present innovative optimization frameworks alongside speed enhancement methods to boost retrieval performance throughout various datasets. Developers target three main goals: lowering memory consumption and enhancing indexing systems, and achieving parallel processing. Table 6 summarizes research papers in the context of scalability and efficiency. Researchers focus on developing multimedia index algorithms to enhance both speed-related capabilities and scalability dimensions and user navigation features. Research by Gareshma et al. [29] developed better audio navigation through speech indexing that used keywords. Graph Code enabled fast information retrieval according to Wagenpfeil et al. [30] but faced challenges due to heterogeneous data sources. The study by Kvet and Papan [31] implemented optimization on B+ Tree indexing strategies to reduce search expenses at the cost of additional operational challenges. Zhu et al. [32] introduced

the DIMS framework that demonstrates malleability, together with a proposal to merge it with learning approaches.

G. REMOTE SENSING IMAGE RETRIEVAL

The retrieval systems for remote sensing images experience major difficulties because of their vast scale and exponential complexity demands. Research efforts for improving retrieval models and scalability of satellite and aerial image systems develop deep learning networks while implementing new hashing methods and distinctive feature extraction approaches. Table 7 summarizes research papers based on the remote sensing image retrieval technique. The field of remote sensing retrieval makes use of deep learning alongside hashing methods to achieve higher accuracy. The application of CCGH by Han et al. [33] combined with multilabel learning and Jaccard reranking provided by Imbriaco et al. [34] attempted to decrease intra-class differences. Deep Attention Hashing, which focuses on features, was introduced by Zhang et al. [35] alongside the fuzzy rule implementation with CNN features described by Ye et al. [62]. Despite these limitations, all systems proved effective with higher mAP performance metrics.

H. MULTIMODAL INTEGRATION AND CROSS-MODAL RETRIEVAL

The task of combining multiple modalities while building efficient cross-modal retrieval systems presents substantial difficulties. Studies have focused on developing structures that enable the proper alignment of different data formats from images through to text and audio and video content for better retrieval performance. Table 8 summarizes research papers based on multimodal integration and cross-modal retrieval techniques. The field of cross-modal retrieval has advanced through deep learning combined with semantic alignment methods. Text-image retrieval gets an enhancement from CCA according to research by Rasiwasia et al. [14]. Better audio-video retrieval became possible through the use of deep networks according to Sun et al. [36], yet these methods lack real-time application capabilities. The strong but computationally intense multi-view diffusion model was proposed by Li et al. [37]. Image-text retrieval performance was enhanced using transformers by Liu et al. [38] while the system required intense computational resources.

I. DATABASE INDEXING ENHANCEMENTS

To achieve scalable retrieval systems with high-performance database indexing proves essential. Current research in database indexing centers on developing B+ Trees through enhanced technology while adding machine learning capabilities and supporting flexible operations. Table 9 summarizes research papers based on database indexing techniques. The method presented by Kvet and Papan [31] enhanced B+ Trees to achieve higher retrieval efficiency, which substantially decreased costs at the expense of increased overhead. The work of Maharjan et al. [39] combined B+-trees with

learned indexes to speed up database operations, but their effectiveness required recent retraining when workloads become dynamic.

J. SPECIALIZED RETRIEVAL SYSTEMS

Special retrieval systems exist for distinct utilization scenarios, including the Metaverse as well as advanced audio-video retrieval systems and dedicated systems for unique environments. Novel architecture combines with learning techniques in specialized retrieval systems to enhance both user interaction quality and retrieval system expandability and efficiency. Specialized systems are tailored for unique applications such as medical imaging, remote sensing, and multimodal data retrieval. Table 10 summarizes research papers based on Specialized cases. Different techniques have improved multimedia retrieval methods during recent research periods. Wagenpfeil et al. [30] improved the speed of retrieval when they adopted Graph Code encoding. Steinert et al. [40] integrated MMIR with the Metaverse for interactive analysis. The combination of hybrid models in water quality prediction led to high prediction accuracy according to Aslam et al. [41]. The combination of object tracking with visual transformers led to better annotation accuracy according to Fernandes et al. [42]. Research by Arslan et al. [43] incorporated deep learning with handcrafted features for image retrieval tasks, although this approach required significant computational resources.

K. AUDIO AND VIDEO RETRIEVAL SYSTEMS

Video and audio retrieval systems enhance efficiency and accuracy by implementing new indexing and extraction features that integrate multiple media forms. Research directs its attention to developing multimedia retrieval systems by improving three areas: system robustness, versatility, and user interaction capabilities. Table 11 summarizes research papers based on Video and Audio retrieval systems. Gareshma et al. [29] improved navigation through speech recognition enhancements for audio indexing until they faced challenges with background noise. The search performance of audio-video retrieval experienced improvements with VGG-based embeddings, according to Sun [36], yet encountered challenges due to high processing requirements. Yang et al. [44] accomplished high surgical video retrieval accuracy using semantic hashing, yet their approach requires improved automatic segmentation.

L. RECENT ADVANCES IN DEEP AND COMPOSITIONAL MULTIMODAL RETRIEVAL

In recent years, multimodal retrieval has progressed from shallow correlation techniques to highly expressive models that support compositional reasoning, domain adaptation, and task-conditioned interaction. A new wave of research focuses on leveraging large-scale pretrained models, prompt-based alignment, and cognitive modeling to enhance semantic retrieval and user intent satisfaction.

TABLE 1. Summary of research papers based on CBIR.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations	Conclusions
Ritendra Datta et al. [3]	2008	CBIR evolution using machine learning to reduce semantic gap.	Visual signatures, relevance feedback, and interaction.	Handcrafted features; no DL used.	Improved feature extraction and accuracy.	Semantic gap persists.	Interdisciplinary CBIR efforts needed.
Ashraf et al. [9]	2020	Fusion of low-level features improves CBIR.	Color Moments, Gabor, DWT, CEDD.	Gabor = 5×5, DWT = 2, CEDD = 144.	High precision/recall.	Handcrafted features limit performance.	Fusion helps; ML can boost further.
D. Srivastava et al. [10]	2023	Deep + local/global features boost CBIR.	CNN-based method.	Dataset: COCO subset.	Better accuracy.	Noisy images, high compute cost.	Recommends deep features.
J. Pradhan et al. [11]	2023	DNA encoding reduces feature dimensions.	DNA → amino acids → SVM/RF.	Sequence length: 200–300.	High precision/recall.	Heavy resource usage.	Suggests combining with DL.
K. N. Sukhia et al. [12]	2022	Dense angle descriptors with dictionary learning.	HOG, OMP, pyramids.	Cell = 8×8, dictionary = 512, pyramid = 3.	Robust to scale and rotation.	Less effective on textured data.	Future work on reducing complexity.

TABLE 2. Summary of research papers based on modern retrieval systems.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations	Conclusions
Michael et al. [13]	2006	MIR addresses semantic gaps, extensibility, and multimodal data integration.	Content-based retrieval, relevance feedback, high-performance indexing.	MPEG-7 indexing, SVM classifiers, relevance feedback; no DL used.	Improved accuracy through ML and better interfaces.	Limited semantic detection and slow commercial adoption.	Need for standard benchmarks and advanced fusion techniques.
Snoek et al. [5]	2005	Multimodal indexing improves access via fusion of visual, audio, and textual streams.	Unified indexing with segmentation and pattern recognition.	TRECVID, MPEG-7, MFCC, SVM, GMM fusion.	Higher indexing precision.	Weak alignment and scalability across modalities.	Further work needed on scalable multimodal integration.
Nikhil Rasiwasia et al. [14]	2010	Cross-modal correlation significantly improves retrieval.	Canonical Correlation Analysis, Logistic Regression.	400-D image, 300-D text; MAP = 0.252; Wikipedia + MIR Flickr.	Best MAP in text–image retrieval tasks.	Sensitive to quality of semantic projection.	Abstraction methods are key to cross-modal success.
Maha Mahmood et al. [15]	2019	Neural networks enhance retrieval accuracy across modalities.	VSM, LSI, Curvelet, DWT, neural networks.	LSI = 100 topics, Curvelet + DWT; 2-layer MLP; local dataset.	High precision and low latency.	High cost in real-time deployment.	Deep learning enables next-gen MIR.
I. Bendib et al. [16]	2018	Semantic enrichment improves retrieval from spoken content.	WordNet, linguistic parsing, keyword spotting.	CMU Sphinx, WordNet 3.0, DTW, POS tagging.	Improved segmentation and higher accuracy.	Requires accurate transcription and NLP tools.	Suggest cloud-based and multilingual extension.

For example, Uni-Retrieval [45] proposes a multi-style framework for STEM education retrieval, incorporating a diverse prompt bank and cross-modal representations to enhance precision in domain-specific queries. Similarly, Yang et al. [44] introduce a composition-based query image retrieval system, where a target is retrieved by combining a

reference image with textual modifications. This approach refines the retrieval process by modeling visual-language interactions through example mining. On the cognitive front, Zhang et al. [46] propose the Chimera framework, which fuses aesthetic and cognitive cues to improve multimodal aspect-based sentiment analysis—a

TABLE 3. Summary of research papers based on hashing and indexing techniques.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations	Conclusions
J. Ahmad et al. [17]	2018	Bidirectional FFT creates compact binary hashes from deep features.	Neuronal sensitivity + FFT for hash code generation.	Hash=512; FFT window=128; sensitivity=0.2; CIFAR-10, Corel-1K.	97% retrieval with 512-bit hashes.	Short hash codes yield lower accuracy.	Scalable and data-independent; wavelets may help.
M. Wang et al. [18]	2024	Deep Index with probabilistic quantization improves retrieval.	Codebook-free quantization, reward-based learning.	Hash=64; Adam; LR=0.0005; entropy=0.8; CIFAR-100, NUS-WIDE.	Higher accuracy and semantic partitioning.	Needs tuning for large datasets.	Good for scalability; adaptivity and feedback needed.
K. Jiang et al. [21]	2025	ROH boosts cross-modal retrieval using MED embedding.	Bridging + online learning + MED embedding.	Embedding=128; Hash=64; $\alpha = 0.5$; SGD; MIRFLICKR, Wiki.	Outperforms SOTA in long-hash settings.	Short hashes lower performance; tuning needed.	Real-time scalable retrieval; improve for short hashes.
J.-M. Guo et al. [19]	2023	UDBH retains structure for better image retrieval.	ViT + async learning + double-bit hashing.	ViT-base; Hash=96; AdamW; patch=16; ImageNet-1K.	High mAP across benchmarks.	Resource-heavy training.	Effective; reduce compute needs.
L. Jin et al. [20]	2025	RCSH improves retrieval via self-supervised hashing.	Dual-prototype contrastive and relational consistency loss.	Prototype=20; Hash=48; Batch=128; Margin=0.4; MS-COCO, CIFAR-10.	Better generalization, top scores on datasets.	Imbalance-sensitive; parameter dependent.	Promising for scale; needs robustness enhancements.

direction increasingly relevant to emotionally aware retrieval systems in marketing, social analysis, and personalized search.

Further, broader studies on deep multimodal learning for information retrieval [47] emphasize scalable, end-to-end architectures trained on large unstructured data with contrastive or generative objectives. These models explore domain generalization, compositional semantics, and architectural flexibility using encoder-decoder or unified transformer backbones.

Recent models have also explored retrieval grounded in compositional semantics. The method proposed by Zhang et al. [48] introduces a novel approach that mines aligned visual-textual pairs from unannotated data to match composed queries. This is particularly valuable in tasks where retrieval queries involve multiple interdependent concepts (e.g., “a person riding a white horse at sunset”), improving generalization in open-world settings.

In the domain of task-specific retrieval, recent work has moved beyond general similarity to incorporate affective and cognitive aspects. For instance, *Exploring Cognitive and Aesthetic Causality for Multimodal Aspect-Based Sentiment Analysis* [49] presents a novel architecture that models sentiment causality using hierarchical attention and integrates aesthetic quality for improved alignment between textual opinions and visual content. This model is particularly effective in scenarios such as social media analysis and brand

monitoring, where emotional alignment across modalities is critical.

Table 12 summarizes recent compositional and cognitively enriched multimodal retrieval models. For instance, Uni-Retrieval introduces multi-style prompt tuning and domain-adaptive query expansion to personalize retrieval in educational contexts. Composition Example Mining applies visual-text relation modeling to improve modified-image query understanding, while Deep Multimodal IR fuses contrastive and decoder-based architectures (e.g., ViLT + BLIP) for domain-general retrieval performance. Other models—such as Multimodal Sentiment Causality and Cognitive-Aesthetic Sentiment Analysis—explore hierarchical attention and aesthetic fusion to enrich interpretability and affective alignment. These innovations reflect a growing shift toward task-specific reasoning, compositional semantics, and lightweight adaptation. The technical parameters—such as prompt size, embedding dimensionality, and attention layers—highlight the design decisions that drive trade-offs in retrieval accuracy, domain generalization, and computational cost.

M. REAL-WORLD APPLICATIONS AND DEPLOYMENT SCENARIOS

Multimodal retrieval systems have evolved beyond theoretical constructs, finding practical applications across

TABLE 4. Summary of research papers based on privacy-preserving techniques.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations and Conclusions
Zhang et al. [22]	2020	Enables efficient privacy-preserving image retrieval with dynamic update capability.	Uses extended locality-sensitive hashing (eLSH), secret sharing, and cuckoo hashing.	eLSH = 256 bits; Cuckoo table = 2^{10} ; Share threshold = 3; Dataset = Caltech-256.	Constant-time retrieval, high accuracy, low key overhead.	Limited scalability explored. Suggests multilingual and scalable extensions.
Anyu et al. [23]	2020	Ciphertext-based retrieval using chaos theory achieves strong privacy.	Uses 4-D hyperchaotic system, DPSH hashing, secure kNN, AES key encryption.	DPSH = 64 bits; AES key = 128 bits; Chaotic $\lambda = 3.99$; Dataset = MNIST, CIFAR-10.	mAP = 0.8931 with strong attack resistance.	High computation cost limits scale. Recommends multimodal optimization.
Y. Ma et al. [24]	2024	TPE framework enhances privacy in IoT image retrieval.	Combines TPE with Huffman coding, adaptive encryption, HSV-ULBP descriptors.	Image = 128×128; ULBP radius = 1, neighbors = 8; Dataset = IoT-DIR.	64% mAP with low file expansion and accurate results.	IoT generalization and scalability remain open issues. Suggest adaptive tuning.

TABLE 5. Summary of research papers based on machine learning and deep learning-based approaches.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations and Conclusions
Liu et al. [25]	2020	ABAC framework improves policy-based retrieval efficiency.	Uses binary identifiers with decision-tree filtering for access policies.	ResNet-18; batch = 64; LR = 0.001; step = 0.1; Dataset = UCMerced, AID.	3–5× faster than traditional policy-based methods.	Limited adaptability in evolving contexts; recommends integrating encryption.
Mahmood et al. [15]	2019	Neural networks enhance multimodal image retrieval.	Combines VSM, LSI, Curvelet + DWT preprocessing, and neural features.	OML margin = 0.25; FV = 2048; Dataset = BigEarthNet19, ML-AID.	High precision and improved retrieval speed.	Scalability remains a challenge due to compute demand.
Rateria et al. [26]	2024	Context-aware models improve semantic relevance in retrieval.	Uses BERT embeddings, WMD, and FSM for query processing.	Hash = 64 bits; LR = 0.0001; attention = 64; Dataset = PatternNet, AID.	Outperforms traditional ranking methods.	Struggles with unseen terms; needs scalable architecture.
Hambarde et al. [27]	2023	Transformer hybrids improve indexing and retrieval.	Surveyed SPLADE, ColBERT, and dense-sparse fusion techniques.	Qualitative; no explicit hyperparameters; Datasets = MS MARCO, SQuAD, WebQA.	Better ranking precision in real-world benchmarks.	High data and compute requirements; future focus: multimodal IR.
Chen et al. [28]	2023	Combining CNNs and transformers boosts retrieval performance.	Uses NetVLAD + GeM pooling over CNN (VGG16) and fine-tuning.	CNN = VGG16; fuzzy threshold = 0.7; margin = 0.1; Dataset = UCMD, PatternNet.	Achieved SOTA performance.	Efficiency vs. generalization remains an open trade-off.

diverse domains. The following examples illustrate how these systems are being deployed to address real-world challenges:

- **Healthcare and Surgical Informatics:** The Semantic-Preserving Surgical Video Retrieval (SPSVR) framework enhances the retrieval of surgical videos by incorporating phase and behavior semantics. Systems like BLIP-2 and MiniGPT-4 can also be adapted to medical domains by aligning surgical descriptions with procedural video frames, enabling explainable retrieval and context-aware content indexing, which are vital for clinical training and intraoperative support [44].
- **Remote Sensing and Environmental Monitoring:** Deep Attention Hashing (DAH) with Distance-Adaptive Ranking improves remote sensing image retrieval by

- emphasizing salient visual features. These techniques, when combined with cross-modal models such as ALIGN, can integrate text-based queries (e.g., “deforested zones in rainy regions”) with satellite imagery, enhancing environmental monitoring tasks such as disaster response or land use classification [46].
- **Education and E-Learning:** Uni-Retrieval is a multi-style retrieval framework tailored for STEM education. It leverages diverse query formats—textual, visual, or problem-based—and a continuously evolving prompt bank. Integration with vision-language models like CLIP or Kosmos-1 supports conceptual grounding across diagrams, lectures, and formulae, thereby enhancing personalized and multimodal learning experiences [47].

TABLE 6. Summary of research papers based on scalability and efficiency.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations and Conclusions
Gareshma et al. [29]	2022	Audio indexing enhances multimedia navigation.	Uses PocketSphinx for speech recognition and keyword spotting.	Binary ID = 64 bits; Decision tree depth = 5; 10K ABAC policies; PocketSphinx API.	Improved navigation using timestamp-based keyword alignment.	Limited to English; noise sensitivity. Future work: ontology-driven clustering.
Wagenpfeil et al. [30]	2021	Graph Code representation improves retrieval latency.	Uses graph encoding, valuation matrices, and similarity metrics.	Curvelet = 5 scales; DWT = 2; NN = 3 layers; TF-IDF; Dataset = Flickr25K.	Achieved 10–60ms retrieval time with high precision.	Requires consistent features; struggles with data heterogeneity. Recommends fusion strategies.
Kvet et al. [31]	2022	Optimized B+Tree indexing reduces search cost.	Autonomous balancing and NULL value handling in B+Tree index.	B+Tree; WMD = 0.7; FSM = 5; BERT base embeddings; Custom QA Corpus.	Reduced retrieval cost by 97%; improved transaction efficiency.	High update overhead; memory use needs reduction.
Y. Zhu et al. [32]	2025	DIMS enables scalable multimodal similarity search.	Heterogeneous partitioning with cost-based optimizer.	ResNet-50; NetVLAD with 64 clusters; GeM pooling (p=3.0); Dataset = Oxford5K, Paris6K.	Strong scalability and faster similarity match.	Needs ML-based indexing and approximation. Future: deep indexing integration.

TABLE 7. Summary of research papers based on remote sensing image retrieval technique.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations and Conclusions
L. Han et al. [33]	2023	CCGH framework reduces intra-class variations and enhances retrieval.	Combines ResNet-18 with cohesion loss and gradual sign activation.	Backbone = ResNet-18; Activation = GSL; Datasets = UCMerced, AID.	Achieves high mAP and rapid convergence.	Struggles with scalability; needs better efficiency tuning.
R. Imbriaco et al. [34]	2022	Multilabel-aware retrieval improves generalization and accuracy.	Uses OML loss with Jaccard-based reranking.	OML loss; Jaccard affinity; Datasets = DLRSD, ML-AID, WHDL, BigEarthNet19.	Outperforms baselines on mAP and nDCG metrics.	Suffers under class imbalance; high computational demand.
Y. Zhang et al. [35]	2023	DAH model captures salient features using attention fusion.	Combines joint channel–spatial attention with balanced weighted loss.	Attention = Channel–Spatial; Loss = Balanced Pairwise; Datasets = AID, PatternNet.	Delivers high mAP and low ANMRR.	Sensitive to object complexity and high compute costs. Recommends multilabel tuning.
F. Ye et al. [62]	2022	Fuzzy rule-based CNN improves classification-driven retrieval.	Fuses fuzzy class membership rules with CNN feature extraction.	CNN descriptor; fuzzy confidence scoring; Datasets = UCMD, PatternNet.	mAP improvement: 90.80%→96.54% (UCMD), 96.58%→98.28% (PatternNet).	Model relies on CNN quality and struggles with class imbalance. Suggests adaptability enhancement.

- **Smart Cities and IoT:** Privacy-aware retrieval systems such as PicAlert are vital for managing sensitive images in smart surveillance. By incorporating privacy-preserving hashing (e.g., UDBH, ROH) and lightweight transformer variants, these systems can operate in real-time at the edge, handling cross-modal queries (e.g., “child running with sound alert near exit A”) with semantic accuracy and speed, which is essential for scalable smart city infrastructure [50].
- **Cultural Heritage and Digital Archives:** Compositional image retrieval techniques have been effectively applied to digital archives, allowing semantic and spatial reasoning over historical artifacts. Emerging

systems can pair textual metadata with visual content using multimodal fusion models to retrieve culturally significant items, facilitating more nuanced archival exploration and educational dissemination [67].

These examples underscore the versatility and impact of multimodal retrieval systems in addressing domain-specific challenges, highlighting their significance in practical applications.

N. FAIRNESS AND EXPLAINABILITY IN RETRIEVAL

Seminal explainable AI methods such as LIME (Local Interpretable Model-Agnostic Explanations) by Ribeiro et al. [51]

TABLE 8. Summary of research papers based on multimodal integration and cross-modal retrieval techniques.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations
Nikhil Rasiwasia et al. [14]	2010	Cross-modal correlations improve retrieval for image–text tasks.	Uses CCA, Logistic Regression, and Semantic Correlation Matching.	Projection dim = 300; MAP metric; Dataset = Wikipedia Retrieval Benchmark.	Achieved best MAP score of 0.252.	Relies heavily on semantic alignment quality and dataset coverage.
M. Sun et al. [36]	2024	Deep learning improves audio–video retrieval fidelity and compression.	Uses VGG, SKM, and cross-learning embeddings.	VGG depth = 16; SKM $\alpha = 0.1$; Embedding dim = 512; Datasets = AudioSet, YouTube-8M.	30.75% boost in fidelity; 7.09% gain in compression.	High GPU demand; lacks real-time responsiveness.
Q. Li et al. [37]	2023	MVD framework enhances multi-view fusion for robust retrieval.	Integrates weight learning, diffusion, and graph-based optimization.	Views = 3; Steps = 5; Tolerance = 1e-3; Datasets = Pascal VOC, NUS-WIDE.	Outperforms 13/16 baselines on standard benchmarks.	Computationally intensive; needs careful hyperparameter tuning.
Z. Liu et al. [38]	2023	CSIC model strengthens semantic alignment for image–text retrieval.	Uses self-attention, UNITER backbone, and transformer blocks.	UNITER base; Batch = 32; Heads = 12; LR = 2e-5; Datasets = Flickr30K, MS-COCO.	Improves Recall@K for both retrieval directions.	Requires region-based detection and large compute.

TABLE 9. Summary of research papers based on database indexing techniques.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations & Conclusions
Kvet et al. [31]	2022	Improved indexing performance through enhanced B+Tree operations and NULL management.	Combines B+Tree optimization with autonomous transaction handling.	BERT base uncased; FSM states = 5; Dataset: Domain-specific QA corpus.	Achieved 97% reduction in retrieval cost with better transaction handling.	Overhead remains for dynamic updates. Future focus: reduce memory footprint and boost adaptability.
S. Maharjan et al. [39]	2023	Learned Store integrates ML-based indexes with B+-trees for better performance.	Switches between learned models and B+-trees using key prediction heuristics.	Model: 2-layer NN; Input: key histograms; Dataset: YCSB-A.	2.29× throughput gain, 6.84× reduction in tail latency.	Needs retraining under high-write conditions. Further work: improve adaptability under load.

and SHAP (SHapley Additive exPlanations) by Lundberg et al. [52] have emerged as standard tools for interpreting complex models. These techniques offer post-hoc interpretability by approximating a model’s local behavior or attributing predictions to input features, respectively. In the context of multimodal retrieval, integrating LIME or SHAP with embedding or ranking layers can help trace the influence of specific visual or textual tokens on the retrieval outcome. This is particularly critical in domains like healthcare, education, or surveillance, where model accountability, transparency, and user trust are paramount.

While the literature reviewed thus far outlines critical developments in content-based retrieval, cross-modal indexing, deep learning, and privacy-enhanced architectures, much of the existing work remains constrained to low- or mid-level feature alignment, task-specific generalization, and unimodal or bimodal data streams. Moreover, limitations in

versatility, semantic fusion, and robustness under dynamic data conditions persist across the surveyed methods.

Recently, however, the research landscape has witnessed a shift toward large-scale pretrained architectures that unify visual and linguistic modalities using instruction tuning, contrastive learning, and generative reasoning. These transformer-based systems—discussed in the next section—present a new design paradigm that integrates cross-modal understanding with few-shot adaptability and real-time interaction. III builds upon the foundational models reviewed above to provide a comparative and critical synthesis of these emerging architectures, their trade-offs, and their implications for next-generation retrieval systems.

III. COMPARATIVE ANALYSIS AND CRITICAL SYNTHESIS

The progression of multimodal multimedia retrieval systems over the past two decades reveals a clear methodological and

TABLE 10. Summary of research papers based on specialized techniques.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations & Conclusions
Wagenpfeil et al. [30]	2021	Graph Code improves multimedia retrieval speed via structured encoding and similarity metrics.	Graph Code En-coding, Valuation Matrices, Similar-ity Metrics.	Curvelet scale=5; DWT=2; NN layers=3; VSM=TF-IDF; Dataset=Flickr25K.	10–60ms retrieval latency with high precision and re-call.	Requires consistent feature extraction; struggles with heterogeneous data. Suggests feature fusion and multilingual support.
Steinert et al. [40]	2023	MMIR–Metaverse integration enhances interactivity and multimedia analysis.	GMIA4MP architecture, PFMR framework.	Multimedia stream=1.2GB/hr; Benchmarked on 3 VR platforms.	Effective integra-tion with meta-verse; good inter-activity.	Lack of standardized datasets and complex environments. Future work: scalability and standard benchmarks.
B. Aslam et al. [41]	2022	Hybrid models improve environmental prediction accuracy.	Random Trees, Random Forest, M5P, REPT, hybrid RT-ANN.	N=1880; Train/Test=70:30; LR=0.01; Epochs=100.	NSE=0.945, RMSE=2.319.	Region-specific data hinders generalizability. Suggests diversified datasets.
R. Fernandes et al. [42]	2024	Object tracking + image retrieval improves annotation efficiency.	CNNs, Visual Transformers (ViTs), object tracking.	ViT=Base-16; Samples=10k; FPS=25; Dataset=MS-COCO.	Improved detection accuracy and annotation speed.	High compute cost; less robust in dynamic settings. Future focus on real-time adaptability.
M. Arslan et al. [43]	2025	Deep + handcrafted features enhance image retrieval.	CNNs (VGG19, InceptionV3, AlexNet), grayscale, BoW indexing.	Model=VGG19; Dataset=CIFAR-10; BoW clusters=1000; Res=224×224.	MAP=95% with VGG19.	High computational load; weak on unstructured data. Future work should target real-world optimization.

architectural evolution, from early feature-based techniques to unified transformer-based architectures. Early models such as Snoek et al. and Rasiwasia et al. leveraged low-level content features and statistical correlation (e.g., CCA) for cross-modal alignment. However, the emergence of contrastive learning (CLIP, 2021), cross-modal prompting (Kosmos-1, 2023), and generative decoding (GIT2, 2023) has reshaped how vision and language modalities are integrated, interpreted, and retrieved.

To highlight this shift, Table 13 presents a model evolution overview across five technical eras—Classical, Early Multi-modal, Contrastive Pretraining, Few-Shot Vision-Language, and Generative Multimodal Modeling. These tools offer a structured backdrop against which contemporary models can be compared in terms of fusion strategy, modality coverage, training paradigms, and trade-offs.

Building on this progression, the next section presents a comparative synthesis of recent vision-language models such

as CLIP, ALIGN, BLIP-2, Flamingo, and MiniGPT-4, focusing on their architectural distinctions, fusion mechanisms, evaluation metrics, and current limitations.

A. DIVERGING DESIGN STRATEGIES

Multimodal systems employ a variety of design paradigms depending on their retrieval objectives:

- i. CLIP and ALIGN use contrastive learning with dual encoders for efficient, zero-shot text-image retrieval, but suffer from shallow alignment.
- ii. BLIP-2 and Flamingo introduce decoder-based and few-shot strategies with stronger visual grounding but higher compute costs.
- iii. MiniGPT-4 represents a conversational, instruction-tuned multimodal system suitable for real-time interactive search.

These models differ not only in architecture but also in their data scale, training objectives, and modality coverage.

TABLE 11. Summary of research papers based on audio and video retrieval systems.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations & Conclusions
Gareshma et al. [29]	2022	Audio indexing tool improves multimedia navigation using keyword spotting.	PocketSphinx speech recognition, DT-based policy, hashing.	Binary ID=64; DT depth=5; ABAC policy=10k.	Enables keyword-timestamp navigation.	Limited robustness to noise; lacks language flexibility. Suggests broader language support.
M. Sun et al. [36]	2024	Deep learning enhances audio-video retrieval via advanced embedding and keyframe clustering.	VGG backbone, adaptive clustering, cross-learning embeddings.	VGG; Keyframe window=5s; Embedding size=512; Dataset=Multimedia	30.75% fidelity gain; 7.09% better compression ratio. AVSet.	Requires high compute; lacks real-time capability. Needs runtime optimization.
Yang Y et al. [44]	2024	SPSVR improves surgical video retrieval with high clinical relevance.	Two-stream features, dual-level hashing, self-supervised learning.	Hash=64 bits; Frame rate=30fps; ResNet-50; Dataset=Cholec80.	Clinically validated with high retrieval accuracy.	Relies on pre-segmented videos. Future work: automatic phase segmentation.

TABLE 12. Summary of recent deep and compositional multimodal retrieval models.

Paper	Year	Key Findings	Method	Technical Parameters	Results	Limitations & Conclusions
Uni-Retrieval	2023	Prompt tuning enhances retrieval in STEM-focused multimodal search.	Adaptive prompt banks with CLIP backbone.	Prompt size=128 tokens; Backbone=CLIP; Dataset=EduVL.	Higher recall and precision over vision-language baselines.	Limited to STEM domains; requires well-curated prompts.
Composition Example Mining	2023	Enhances compositional image-text understanding.	Visual-text composition mining with relation modeling.	Relation net=3 layers; Embedding=512; Dataset=COCO+VG.	Stronger performance on modified-image queries.	Limited generalization; quality of examples impacts performance.
Multimodal Sentiment Causality	2022	Cross-modal causal reasoning improves sentiment analysis.	Attention fusion using cognitive-aesthetic features.	Heads=8; Audio rate=16kHz; Dataset=MOSI, MOSEI.	More precise predictions with causal explanation.	High model complexity; useful for interpretability.
Deep Multimodal IR	2022	Transformers improve retrieval across multimodal datasets.	Contrastive and decoder-based transformer fusion.	ViLT+BLIP; Batch=64; Dataset=Flickr30K, MS-COCO.	State-of-the-art results on standard benchmarks.	Requires optimization for efficient deployment.
Composition Example Mining (Unsupervised)	2022	Enables open-world generalization using mined compositions.	Learning from unannotated image-text pairs.	Embedding=512; Transformer=CLIP-based; Dataset=COCO+VG.	High recall/precision on composed queries.	No explicit supervision; performance depends on data variety.
Cognitive-Aesthetic Causality for MABSA	2023	Attention-based fusion enhances sentiment-image alignment.	Multimodal attention with lexical-aesthetic fusion.	Layers=2–3; CNN aesthetics=VGG; Dataset=MABSA.	Outperforms sentiment baselines with improved coherence.	Domain-specific; less applicable to generic retrieval tasks.

Table 16 offers a comparative analysis of model architectures across key performance dimensions. BLIP-2 employs a two-stage decoder over a ViT-G backbone (80B parameters) to support few-shot generalization, whereas MiniGPT-4 fine-tunes a Q-Former atop a frozen encoder to enable lightweight interactive retrieval. Kosmos-1 integrates instruction-tuned prompting with cross-modal reasoning over text, audio, and image data. Privacy-aware systems

(e.g., TPE + Privacy IR) and deep hashing models provide fast, secure retrieval for edge environments, though they often trade off semantic precision.

Across these models, a recurring tension emerges between representational depth and latency. Transformer-heavy models like BLIP-2 and Flamingo achieve high accuracy (85–88) but demand significant GPU resources, limiting deployment. In contrast, hash-based systems enable real-time retrieval with compact representations, making them suitable for IoT or surveillance. Uni-Retrieval offers a middle path, combining prompt banks with adaptive retrieval for low-resource, domain-specific tasks. These architectural trade-offs—governed by task complexity, modality richness, and resource constraints—underscore the need for standardized benchmarking to ensure reproducible, fair evaluation across deployment scenarios.

B. FUSION STRATEGY OPTIMIZATION

These patterns imply important future directions for research and engineering:

- **Hybrid Fusion Mechanisms:** Future models should adopt hybrid fusion mechanisms that balance scalability with fine-grained alignment—e.g., shallow contrastive pretraining with deep attention-based refinement.
- **Multimodality Inclusion:** There is a need to go beyond static image-text models and incorporate audio, temporal, and sensory streams, particularly for surveillance, robotics, and assistive technologies.
- **Dynamic Adaptability:** Models should be capable of few-shot adaptation or continual learning, allowing them to incorporate user feedback and evolving context without full retraining.
- **Explainability and Trust:** Incorporating interpretability mechanisms (like attention maps or attribution scores) is vital, especially for domains like healthcare, legal retrieval, and surveillance.

C. ANALYTICAL OBSERVATIONS AND THEMATIC INSIGHTS

Building upon the comparative analysis in Table 14, this section distills recurring themes and systemic limitations in current multimodal multimedia retrieval models.

- i. **Shallow vs. Deep Fusion Trade-offs:** Models employing shallow fusion (e.g., CLIP, ALIGN) prioritize computational efficiency but fail at fine-grained semantic relationships. In contrast, deep fusion models (e.g., BLIP-2, Flamingo) offer stronger alignment but incur high computational costs.
- ii. **Modality Narrowness:** Most systems are limited to image-text pairs. Few (e.g., Flamingo, Kosmos-1) incorporate video or audio modalities, limiting generalizability.
- iii. **Task-Specific Finetuning vs. Generalization:** Instruction-tuned models (e.g., MiniGPT-4, BLIP-2) perform well on curated datasets but generalize poorly to noisy or domain-shifted data.

- iv. **Scaling Cost vs. Performance:** Models like Flamingo demonstrate strong few-shot performance but require 80B+ parameters. In contrast, smaller models (e.g., MiniGPT-4) approach comparable performance at lower resource costs.
- v. **Missing Interpretability:** Most systems lack robust methods for explaining retrieval outcomes, limiting trust and adoption in sensitive domains.

D. DESIGN IMPLICATIONS FOR NEXT-GENERATION RETRIEVAL SYSTEMS

Building on the comparative analysis and architectural trade-offs, we propose grounded design principles:

- **Modality-Enriched Fusion Architectures:** Future systems should incorporate temporal, audio, and sensor streams for applications like surveillance and healthcare.
- **Efficient Semantic Alignment:** Lightweight alignment techniques, such as adapter layers and hybrid attention, should be prioritized for deployment on edge devices.
- **Interactive and Continual Adaptation:** Systems should integrate mechanisms for continual learning and user feedback without full retraining.
- **Transparent Retrieval Justification:** Explainability via attention visualization, attribution maps, and trace-backs is critical in sensitive domains.
- **Benchmark Unification and Rigor:** Future research should standardize composite benchmarks (e.g., COCO, VQAv2, VizWiz, YouTube-8M) and include interpretability, latency, and robustness metrics.
- **Neuro-Symbolic Approaches:** Combining neural models with symbolic reasoning and knowledge graphs offers improved explainability and generalization.

E. EDGE-COMPATIBLE RETRIEVAL

The trade-off between computational efficiency and accuracy is central to edge-compatible retrieval:

- **DIMS (Deep Indexing for Multimedia Search):** Uses deep feature hashing and cluster-based ANN indexing. Optimized for large-scale multimedia retrieval in bandwidth-constrained settings.
- **Reinforced Object Hashing (ROH):** Employs reinforcement learning to generate compact, semantic-preserving hash codes. Balances precision and memory usage, but faces generalization challenges.

While ROH achieves semantic fidelity with reduced storage, it requires careful design and may not generalize across domains. DIMS offers speed but sacrifices interpretability and adaptability. Complementary strategies—such as quantization, distillation, and low-rank compression—are also being explored, though they often compromise performance in instruction-tuned scenarios.

These architectural insights provide a foundation for identifying open research challenges and inspire the design of future multimodal retrieval systems that are efficient, interpretable, and adaptable to real-world use cases.

IV. A NOVEL CONCEPTUAL TAXONOMY OF MULTIMODAL RETRIEVAL ARCHITECTURES

To support a more unified and analytical understanding of evolving multimodal retrieval paradigms, we propose a novel five-dimensional taxonomy that categorizes recent architectures along critical design axes. This taxonomy is informed by an extensive synthesis of recent state-of-the-art transformer-based and hybrid systems, including BLIP-2, Kosmos, MiniGPT, and Uni-Retrieval. Each axis captures a key technical decision point in system design, offering a scaffold for comparative evaluation and domain-specific adaptation.

A. SIGNIFICANCE AND IMPLICATIONS

This taxonomy performs two crucial roles:

- 1) Architectural Clarity: It classifies emerging systems into interpretable categories, bridging conceptual silos between information retrieval, deep learning, and multimodal representation learning.
- 2) Comparative Benchmarking: It enables researchers and practitioners to evaluate system design choices across datasets, application domains, and computational trade-offs.

In contrast to prior surveys that focus only on technical capabilities or modality scope, our framework emphasizes compositional generalization, query expressiveness, and scalability—key metrics for next-generation multimedia retrieval. Table 13 gives a summary of the Conceptual Taxonomy for Deep Multimodal Retrieval Architectures

B. FUSION STRATEGY

The fusion strategy defines how different modalities—such as image, text, and audio—are integrated within the retrieval pipeline. Four dominant strategies are identified:

- Early Fusion integrates raw modality features before encoding.
Example: Flamingo
- Late Fusion fuses independently learned modality-specific representations at a higher abstraction level.
Example: BLIP-2
- Cross-Modal Interaction uses attention-based mechanisms to capture inter-modal relationships dynamically.
Example: CLIP
- Decoder-Based Generation uses decoders to generate outputs (e.g., captions or responses) conditioned on cross-modal embeddings.
Example: MiniGPT-4

Implication: Fusion choice directly influences retrieval accuracy, latency, and model interpretability.

C. QUERY ADAPTATION

Query adaptation refers to methods that enhance the semantic alignment between user queries and indexed multimodal content. Key variants include:

- Prompt-Tuned: Uses trainable or predefined prompts to condition the model.
Example: Kosmos
- Template-Based: Employs rule-based query structures to inject domain constraints.
Example: Uni-Retrieval
- Latent-Mapped: Maps queries into a shared embedding space through learned transformations.
Example: MiniGPT-4
- Visual Grounding: Links textual query components to spatial or visual elements in content.

Implication: Robust query adaptation enhances retrieval relevance, especially for complex or abstract queries.

D. RETRIEVAL MECHANISM

This dimension captures the core algorithm used to match queries with relevant content:

- Contrastive Learning: Aligns positive pairs and separates negative ones in the shared space.
Example: CSIC
- Attention Alignment: Applies self- or cross-attention to model inter-modal relevance.
- Semantic Hashing: Converts feature vectors into compact binary codes for efficient retrieval.
Example: UDBH
- Instruction Tuning: Modifies behavior using task-specific natural language commands.
Example: Deep Multimodal IR

Implication: Retrieval mechanisms dictate scalability, retrieval quality, and adaptability across domains.

E. SCALABILITY DESIGN

Scalability design addresses how efficiently the retrieval system performs as dataset size increases:

- Approximate Nearest Neighbor (ANN): Accelerates search using clustering or locality-sensitive hashing.
Example: DIMS
- Hashing: Enables sub-linear search complexity via binary encoding.
Example: ROH
- Transformer Pruning: Reduces model size and inference time through structured sparsity.
Example: Deep Index
- Online Learning: Supports continual adaptation without retraining from scratch.
Example: SCH-GAN

Implication: Efficient scalability mechanisms are crucial for real-time and large-scale deployment.

F. MODALITIES USED

The final dimension refers to the input-output modalities supported by the retrieval system:

- Text-Image: The most common bimodal pairing for retrieval.

- Text–Audio–Video: Supports complex real-world applications (e.g., surveillance, medical video retrieval).
Example: Multimodal Sentiment Causality
- Multimodal Feedback Loop: Incorporates iterative refinement through user interaction.
Example: DAH, SPSVR

Implication: Expanding modality support improves system versatility and context comprehension.

This conceptual taxonomy offers a structured framework for understanding, evaluating, and designing deep multimodal retrieval systems. By categorizing recent research across five essential architectural dimensions, it facilitates not only comparative benchmarking but also the identification of under-explored combinations, such as instruction tuning with multimodal feedback—that could serve as fertile ground for future innovation.

However, while this taxonomy highlights the architectural richness and recent advances, it also brings to light several persistent limitations and systemic trade-offs. Many of the reviewed systems, though novel in design, still struggle to meet the practical demands of scalability, semantic alignment, explainability, and robustness. These shortcomings highlight the need for a more in-depth investigation into the unresolved challenges at the intersection of multimodal learning and large-scale information retrieval.

V. RESEARCH GAPS

Despite the substantial progress in multimodal retrieval system design—reflected in diverse fusion strategies, scalable architectures, and sophisticated query adaptation—several critical research challenges remain unresolved. These limitations are not isolated; rather, they are tightly interwoven with the five architectural dimensions outlined in the previous section. For instance, while contrastive learning and cross-modal attention have improved alignment in shared embedding spaces, they often fall short in bridging the semantic gap across heterogeneous modalities. Similarly, although ANN and transformer pruning address scalability, they sometimes compromise retrieval precision or adaptability in dynamic environments.

The core research gaps can be broadly grouped into the following persistent challenges:

- the semantic gap between low-level features and high-level intent,
- retrieval inefficiency in large-scale and real-time scenarios,
- data privacy and robustness issues in cloud and IoT contexts,
- imbalanced or noisy training data, and
- a lack of explainability and trustworthiness in high-stakes applications.

These issues not only limit retrieval effectiveness but also hinder user trust, domain adaptability, and long-term system viability. Identification and filling of these gaps are essential

for the development of next-generation multimedia retrieval systems that are robust, interpretable, and context-aware.

Table 15 summarizes the key research gaps, supported by recent studies, to highlight areas that demand further exploration.

Filling these knowledge gaps is necessary for creating scalable, secure, and advanced approaches for multimedia retrieval. Among these challenges, one trade-off recurs across architectures and applications—the tension between computational efficiency and retrieval accuracy. This trade-off has direct implications for system design, especially in real-time, resource-constrained, and large-scale deployment scenarios. The next section delves deeper into this issue.

VI. TRADE-OFF BETWEEN COMPUTATIONAL EFFICIENCY AND RETRIEVAL ACCURACY

One of the most persistent and cross-cutting challenges highlighted across the architectural taxonomy and the identified research gaps is the trade-off between computational efficiency and retrieval accuracy. This trade-off influences model architecture, deployment feasibility, and real-world applicability, especially in large-scale and time-sensitive environments.

The trade-off between efficiency and accuracy, as shown in Fig. 1, is a big challenge in multimedia retrieval. CBIR (Content-Based Image Retrieval) is very efficient (85%) since it relies on handcrafted features, but has low accuracy (60%), so it's good only for simple retrieval tasks [3], [9], [11], [12]. Machine Learning-Based Retrieval improves accuracy (70%) through statistical learning, but at the cost of reduced efficiency: it needs more computational resources [53], [57], [59], [60]. Deep Learning-Based Retrieval has the highest accuracy (85%) by using CNNs and transformers, but is computationally expensive (60%), so it is not feasible for real-time applications [10], [15], [24], [35], [43]. Hybrid Models and Cross-Modal Retrieval try to balance both efficiency and accuracy, so good for large-scale and real-time multimedia search [9], [14], [21], [26], [38], [41]. To optimize future retrieval systems, we should focus on lightweight deep learning models, adaptive indexing structures, and hardware-efficient AI techniques to boost both speed and accuracy for practical deployment in large-scale and real-time applications.

VII. BENCHMARK DATASETS AND EVALUATION METRICS

To quantitatively evaluate retrieval systems and navigate the trade-off between efficiency and accuracy, the use of standardized benchmark datasets and robust evaluation metrics is critical. These tools facilitate reproducibility, enable fair comparison of different architectures, and provide insight into real-world generalization capabilities.

Standardized datasets include MS-COCO for image captioning, NUS-WIDE for multimodal retrieval, ImageNet for object classification, Corel-1K for content-based image retrieval, Flickr30K for cross-modal retrieval, and Youtube-8M for large-scale video retrieval; these datasets introduce

TABLE 13. Summary of the conceptual taxonomy for deep multimodal retrieval architectures.

Dimension	Design Variants	Representative Examples
Fusion Strategy	Early fusion, late fusion, cross-modal interaction, decoder-based generation	Flamingo (Early) [63], BLIP-2 (Late) [66], CLIP (Cross-Modal) [45], MiniGPT-4 [64]
Query Adaptation	Prompt-tuned, template-based, latent-mapped, visually grounded	Kosmos [65], Uni-Retrieval [47], MiniGPT-4 [64]
Retrieval Mechanism	Contrastive learning, attention alignment, semantic hashing, instruction tuning	CSIC [38], UDBH [69], SCH-GAN [70], Deep Multimodal IR [71]
Scalability Design	ANN search, hashing, transformer pruning, online learning	ROH [21], DIMS, Deep Index [18], RCSH [20]
Modalities Used	Text–image, text–audio–video, multimodal feedback loop	Multimodal Sentiment Causality [72], DAH [68], SPSVR [44]

TABLE 14. Comparative Trade-offs among vision-language models across modalities, fusion strategies, and performance characteristics.

Model	Modality (I=Image, T=Text, V=Video)	Fusion Type	Training Data	Strengths	Limitations
CLIP [45]	I + T	Contrastive	400M image-text pairs	Highly scalable, robust for open-domain retrieval	Weak in grounding and lacks audio/video integration
Flamingo [63]	I + T + V	Few-shot generation	80B+ tokens from multimodal corpora	Strong few-shot QA, contextual awareness	High inference cost and large compute demands
BLIP-2 [66]	I + T	Query encoder fusion	Web-scale caption data	Follows instructions well, structured input handling	Less scalable than CLIP, performance drops on OOD data
MiniGPT-4 [64]	I + T	GPT adapter with vision encoder	Public open-source datasets	Interactive, conversational visual querying	Struggles with complex compositional reasoning

different challenges and benchmarks for models to be tested. The diverse size, difficulty, and quality of annotation of the data heavily influence the training and evaluation of retrieval models. It must be noted that evaluation methods mainly depend on retrieval tasks. Some common metrics for retrieval effectiveness measurement include Mean Average Precision (mAP), Precision@K, Recall@K, Normalized Discounted Cumulative Gain (NDCG), and Top-1 Accuracy. Table 17 gives a summary of benchmark datasets and evaluation metrics.

The fragmentation of datasets across retrieval studies—such as MS-COCO, EduVL, YouTube-8M, and Flickr30K—creates silos of evaluation that hinder generalization and scalability assessments. As seen in the comparative tables, even top-performing models like Flamingo and BLIP-2 are benchmarked on different datasets using inconsistent metrics, which prevents a unified understanding of retrieval effectiveness. A standardization of benchmarks would not only aid empirical comparison but also help assess the true impact of architectural and parameter-level innovations in a reproducible way.

While benchmarks keep establishing comparability from one retrieval approach to another, problems such as dataset

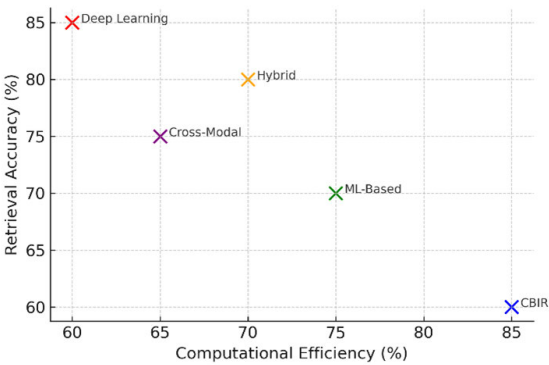


FIGURE 1. Trade-Off between computational efficiency and retrieval accuracy.

biases, changing needs in real-world data, and demands around explainability hinder the making of evaluation frameworks that are quite sturdy and adaptable in their application. The standardization of datasets and metrics of evaluation will keep on playing an important role in making advancement possible to facilitate abstraction, efficiency, and equity in

TABLE 15. Research Gaps: Multimedia information retrieval.

Research Gap	Description	Real-World Implications	Needed Advancement	Related Works
Semantic Alignment Limitations	Difficulty aligning low-level features (e.g., pixels) with high-level semantics.	Causes misinterpretation in sensitive areas like medical imaging, legal search, and surveillance.	Use concept-guided embeddings and attention-based alignment mechanisms.	[3], [13], [6], [14], [9], [10], [43]
Limited Modality Integration	Most models focus on image-text pairs, ignoring video, audio, and sensor data fusion.	Inadequate performance in IoT, smart cities, and robotics applications.	Dynamic modality routing and multimodal fusion of sensor, audio, video, and text.	[14], [15], [56], [21]
Interpretability and Transparency	Closed box deep models provide little insight into retrieval decisions.	Limits trust and usability in high-stakes domains like healthcare or defense.	Incorporate XAI tools: saliency maps, attention heatmaps, attribution analysis.	[15], [10], [27], [38], [28]
Deployment Efficiency	High memory and compute overheads limit real-time deployment.	Infeasible on edge devices or AR/VR setups due to latency.	Apply model compression: quantization, pruning, transformer-lite architectures.	[25], [15], [9], [23], [33], [28], [43]
Robustness to Noisy/Evolving Data	Models fail under imbalance, drift, or user-generated content variance.	Retrieval accuracy drops in dynamic platforms or social media.	Use continual learning, robust loss functions, and adaptive sampling.	[16], [23], [56], [35], [61]
Security and Privacy Constraints	Retrieval systems lack encryption, anonymity, or federated privacy control.	Unsafe for use in smart cities, hospitals, or surveillance cloud services.	Develop secure indexing, federated training, and privacy-preserving pipelines.	[22], [23], [24]
Evaluation & Benchmark Gaps	Fragmented datasets and limited metrics hamper fair assessment.	Difficult to benchmark across tasks, domains, and modalities.	Create unified benchmark suites and expand evaluation metrics (e.g., NDCG, latency).	[13], [5], [6], [40]
Lack of Human-AI Collaboration	Retrieval systems lack interactive correction or learning from feedback.	Reduces adaptability in journalism, legal reviews, and eHealth workflows.	Integrate user-in-the-loop systems and relevance feedback pipelines.	[16], [42]

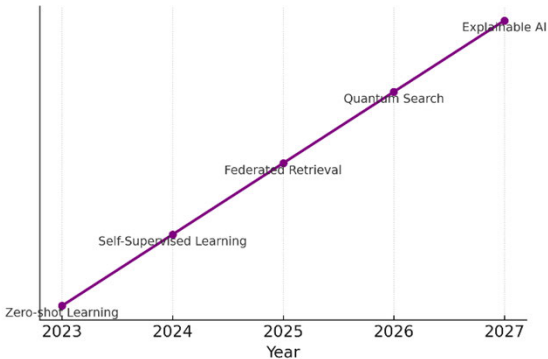


FIGURE 2. Roadmap for future trends in multimedia retrieval.

real-life applications as multimedia retrieval systems extend deeper dating.

VIII. DISCUSSION

A. ARCHITECTURAL TRENDS AND TRADE-OFFS

The comparative landscape synthesized in Tables 12 and 16 illustrates a decisive shift in retrieval system design—from traditional feature-based techniques (e.g., CBIR, linear CCA)

to advanced deep learning and instruction-tuned foundation models (e.g., MiniGPT-4, Kosmos-1). These models offer improved semantic understanding, multi-modality coverage, and few-shot generalization. However, this shift introduces significant trade-offs in terms of compute cost, model complexity, and system interpretability.

Emerging architectures tend to favor late fusion or decoder-based strategies, integrating representations via attention-based layers or prompt conditioning. While this enhances compositionality and alignment, it also increases model size and inference latency. Lightweight models still play a role in edge or privacy-sensitive environments, but often lag in handling complex semantics.

A critical insight from the taxonomy is that no current system achieves optimality across all dimensions—accuracy, scalability, explainability, and real-time performance. This suggests a need for hybrid models that leverage the strengths of both deep and shallow paradigms.

B. LIMITATIONS IN EVALUATION AND BENCHMARK FRAGMENTATION

One of the most pressing limitations across the literature is the lack of consistent, standardized evaluation protocols.

TABLE 16. Comparative analysis of multimodal multimedia retrieval models across key performance dimensions.

Model	Year	Modality	Fusion	Acc. (%)	Technical Parameters	Scale	Explainability	Strengths	Limitations
CBIR [3]	2005	Image	Feature-level	60–75	SIFT, Gabor filters; no deep learning	2	2	Simple and efficient	Lacks semantic reasoning
Deep Hashing [23]	2018	Image–Text	Deep encoding	78–85	Linear CCA; shallow projections	4	2	Fast indexing	Sensitive to noise
TPE + Privacy IR [24]	2020	Image–Text	Secure encoding	~80	CNN encoder; triplet loss; 64-bit hash	2	4	Privacy-preserving	Higher latency
Flamingo [63]	2022	Video–Text	Deep attention	~88	AES + triplet hashing; temporal fusion	2	3	Few-shot, temporal context	High compute cost
BLIP-2 [66]	2023	Image–Text	Two-stage decoder	85–87	ViT-G; 80B parameters; large-scale tuning	3	3	Visual grounding	Inference cost
MiniGPT-4 [64]	2023	Image–Text	Instruction-tuned	85–88	ViT-g + Q-Former; 6B params; frozen encoder	3	3	Real-time, interactive	Domain fragility
Kosmos-1 [65]	2023	Text, Audio, Vision	Prompted transformer	~86	ViT-L + Q-former; LLaVA-tuned	3	3	Multimodal reasoning	Few public benchmarks

As Tables 12 and 16 demonstrate, models are tested across diverse benchmarks (e.g., COCO+VG, EduVL, MABSA, MSR-VTT), each with unique label schemas, modality compositions, and test conditions.

This benchmark fragmentation impairs comparability. Accuracy metrics reported in isolation (e.g., 85–88%) lose significance without consistent cross-dataset evaluation. Further, the absence of unified metrics—such as Recall@K, NDCG, or explainability scores—limits reproducibility and weakens empirical claims. Few studies attempt domain transfer or stress-testing retrieval systems in noisy, real-world environments.

A standardized evaluation suite, akin to GLUE or ImageNet for other subfields, would be instrumental in resolving this issue and enabling more credible benchmarking.

C. EXPLAINABILITY AND HUMAN-CENTRIC GAPS

Despite architectural advances, explainability remains a critical bottleneck. Most state-of-the-art models—especially transformer-based systems—function as closed boxes. They do not provide users with reasons for retrieved results or confidence indicators.

Only a few models (e.g., TPE + Privacy IR) attempt to incorporate traceable encoding or partial interpretability. Yet, these efforts are not widespread. This is especially problematic in high-stakes domains like medicine, law, or education, where opaque results are unacceptable.

Additionally, there is limited progress in incorporating user feedback, personalization, or dynamic re-ranking based on

evolving query intent. Interactive retrieval, active learning loops, and user-grounded explanation mechanisms remain open areas for innovation.

D. TOWARD PRACTICAL AND SCALABLE DEPLOYMENT

The practical deployment of retrieval models is constrained by their computational footprint. Many high-performing models require large-scale GPU infrastructure, hindering their integration in real-time, mobile, or embedded systems. There is a growing demand for privacy-preserving, energy-efficient, and low-latency retrieval architectures.

Techniques like model pruning, quantization, knowledge distillation, and edge-indexing strategies are underexplored in this context. Bridging the gap between high-accuracy retrieval and real-world deployability is critical for broader adoption.

E. SYNTHESIS AND IMPLICATIONS

The analysis in this paper reveals that while retrieval models are improving in capability and sophistication, their development is uneven across key dimensions. Accuracy gains alone do not translate into usability, fairness, or deployability. There is an urgent need for:

- Multimodal evaluation pipelines with unified metrics,
- Architectures that balance accuracy and transparency,
- Systems that can generalize across domains with minimal retraining,
- Integration of user modeling and feedback mechanisms.

TABLE 17. Benchmark datasets and evaluation metrics.

Dataset	Domain	Size	Modality	Annotation Type	Tasks Enabled	Evaluation Metrics
MS-COCO	Image Captioning	330K images	Image + Text	Object labels, image captions	Captioning, Object Detection	BLEU, mAP
NUS-WIDE	Multimodal Retrieval	270K images	Image + Tags	Multi-label tags	Cross-modal retrieval, Tag prediction	mAP, Precision@K
ImageNet	Object Classification	14M images	Image	Category labels	Classification, Transfer Learning	Top-1, Top-5 Accuracy
Corel-1K	Image Retrieval	1K images	Image	Class labels	CBIR: Content-Based Image Retrieval	Precision, Recall
Flickr30K	Cross-Modal Retrieval	30K images	Image + Text	Descriptive image captions	Text-to-Image, Image-to-Text Retrieval	Recall@K
YouTube-8M	Video Retrieval	8M videos	Video + Tags	Multi-label video tags	Video Classification and Retrieval	mAP, NDCG
VizWiz	Accessibility / VQA	~31K images	Image + Text	Questions from unaware users	Visual QA, Accessible Retrieval	Recall@K, NDCG, JQS
CLIPBench	Vision-Language	~400K prompts	Image + Prompt	Instructional prompts	Prompted retrieval, Few-shot learning	Cosine Similarity, Semantic Alignment
VQAv2	Visual QA	~200K images, ~1.1M Qs	Image + Q&A	Human-generated questions and answers	Visual QA, Retrieval	Accuracy, NDCG, BLEU
SPSVR	Medical / Surgical Retrieval	~5K videos	Video + Text	Procedure descriptions	Surgical Retrieval, Explainability	Recall@K, Grounding Score, Faithfulness

TABLE 18. Cross-domain Synergies: Emerging interdisciplinary contributions to multimodal retrieval.

Paradigm	Application Domain	Role in Retrieval	Benefit
Neuro-Symbolic Learning	Biomedical, Legal Informatics, Education	Enables logic-aware and interpretable semantic matching through symbolic representations fused with neural embeddings	Improves explainability, promotes domain-specific trust
Generative AI (e.g., GPT-4V)	Creative Media, Diagnostics, Educational Platforms	Supports prompt-driven retrieval and response generation by leveraging large multimodal language models	Enhances interactivity and contextual reasoning
Causal Inference Models	Social Media, Policy Analytics, Surveillance Systems	Enables root-cause modeling for understanding user behavior, sentiment drift, or algorithmic bias	Promotes fairness, transparency, and bias mitigation

These gaps define the frontier of multimodal multimedia retrieval research. The next section articulates the Open Research Questions (ORQs) that arise from these limitations and trends.

F. OPEN RESEARCH QUESTIONS

To advance the field beyond current limitations, the following Open Research Questions (ORQs) are proposed. These questions are grounded in the comparative synthesis of models, the taxonomy of architectures, and the deployment challenges discussed earlier.

- 1) **How can retrieval systems achieve robust generalization across domains and modalities without extensive retraining?**
Current models often excel on specific datasets but degrade significantly when exposed to cross-domain inputs or novel modality combinations. Research is needed on transfer learning, domain adaptation, and zero-shot retrieval in real-world, noisy settings.
- 2) **Is it possible to design unified multimodal architectures that support real-time, on-device inference without sacrificing retrieval accuracy?**

Foundation models show promise but are too heavy for deployment at the edge. There is a pressing need for lightweight, task-adaptive architectures that operate efficiently across device classes and network conditions.

3) **What explainability techniques can be adapted or invented for deep multimodal retrieval systems?**

As most modern systems operate as closed boxes, users cannot understand or contest their outputs. Research should explore attention visualization, causal inference explanations, and human-centered feedback loops for transparent retrieval.

4) **Can we develop a universal benchmarking suite to evaluate multimodal retrieval models fairly and reproducibly?**

Benchmark fragmentation undermines comparative analysis. A standardized evaluation protocol—with diverse modalities, annotated ground truth, and unified metrics (e.g., NDCG, Recall@K, F1)—would catalyze progress and accountability.

5) **How can user interaction be made an integral part of retrieval, especially in adaptive and long-term systems?**

Most current models are static and do not incorporate user history, context, or feedback. Exploring reinforcement learning, relevance feedback, and preference modeling could enable systems that evolve with user needs while maintaining fairness and privacy.

6) **What are the best strategies for compressing and indexing semantically rich representations at scale?**

As models encode high-dimensional, temporally contextual embeddings (e.g., via Q-Formers or ViTs), new indexing strategies are required. Efficient storage, retrieval, and update mechanisms—potentially using approximate nearest neighbor methods or compressed embeddings—remain open problems.

These questions point toward a new research agenda that prioritizes scalability, explainability, generalization, and user-centric design. Addressing them is essential to building retrieval systems that are not only technically robust but also socially and operationally viable.

IX. CONCLUSION

As evident from the architectural taxonomy, trade-off analysis, and benchmarking evaluations, modern multimedia retrieval systems have made significant strides by integrating deep learning techniques capable of bridging modality gaps and improving representational power. Yet, despite this progress, critical challenges remain unresolved.

Modern multimedia retrieval systems achieve their progress by using deep learning technology, which unifies different media types alongside improved approaches for storing data. Computer systems are still confronting multiple challenges, such as understanding gaps with humans, along with problems related to versatility, transparent decision-making, and resource optimization. The current

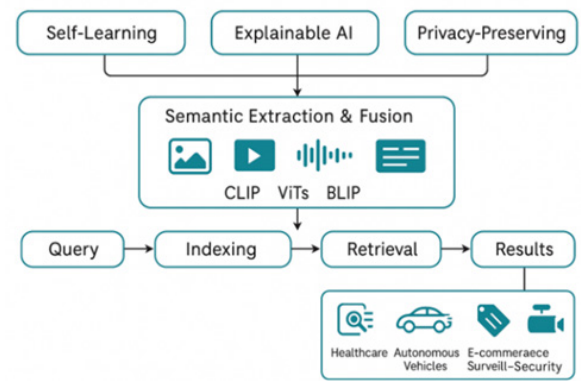


FIGURE 3. Proposed unified architecture.

systems encounter three major problems because they fail to link diverse media types and handle unorganized, multifaceted datasets, and maintain privacy along with security in extensive operations. Further research needs to address the gap through deep learning integration with logical reasoning methods. The two vital requirements for enhanced media retrieval systems comprise stronger data representations along with AI explanations for transparency. The deployment of real-world systems requires better data structure development and basic retrieval systems, together with privacy-preserving mechanisms based on federated learning strategies. Multimedia retrieval systems will become both more scalable and efficient, and easier to use by solving current challenges.

The future development of smart, adaptable, ethical artificial intelligence models will establish better accessibility alongside enhanced usability, mainly within healthcare security and e-commerce domains. Despite substantial algorithmic progress, our review identifies a persistent empirical limitation: benchmark fragmentation. Modern retrieval systems are tested on a diverse set of datasets and performance metrics, which makes cross-model comparisons unreliable. As reflected in Tables 12 and 16, even the most advanced systems vary in how they are evaluated—some emphasize Recall@K or MAP, while others focus on few-shot alignment or latency trade-offs. Anchoring conclusions to standardized evaluation protocols is thus essential.

This review contributes by synthesizing architectural innovations, identifying their empirical trade-offs, and advocating for unified benchmarking practices to guide future research. While the review synthesizes key advances across retrieval methods, it is equally important to ground future work in unified empirical analysis. By highlighting architectural innovations and observed trade-offs, this paper offers a roadmap for selecting suitable retrieval models based on task complexity, latency constraints, and modality integration needs. Further comparative experimentation across standardized benchmarks is essential to validate the scalability, robustness, and domain adaptability of recent models.

X. FUTURE DIRECTIONS

To address the identified research gaps in multimedia retrieval systems, several future research directions are proposed. To overcome persistent challenges in scalability, semantic alignment, and trustworthiness, a unified framework that seamlessly integrates multimodal semantic fusion, explainable AI, privacy-preserving learning, and continual adaptation for robust and future-ready multimedia retrieval is also introduced in this section. These directions aim to improve the accuracy, scalability, interpretability, and security of retrieval systems, enabling more effective and adaptable solutions for diverse and dynamic data environments.

A. BRIDGING THE SEMANTIC GAP IN CONTENT-BASED RETRIEVAL

Future research should focus on developing advanced semantic understanding models that integrate multimodal data, contextual information, and user intent. Hybrid models combining symbolic reasoning with deep learning could enhance semantic alignment between low-level features and high-level concepts, improving retrieval accuracy.

B. ADVANCING CROSS-MODAL RETRIEVAL CAPABILITIES

Efforts should be directed toward improving cross-modal retrieval by developing robust embedding spaces that effectively align heterogeneous data types (text, images, audio, and video). Techniques such as contrastive learning and cross-modal transformers could bridge the modality gap and enable seamless data integration.

C. ENHANCING INTERPRETABILITY AND TRANSPARENCY OF DEEP MODELS

Developing explainable and transparent deep learning models is critical for user trust and adoption. Incorporating explainable AI (XAI) techniques, attention visualization, and model interpretability frameworks can help users understand the decision-making process of retrieval systems, particularly in sensitive domains like healthcare and security.

D. SCALABLE AND EFFICIENT RETRIEVAL FOR LARGE-SCALE DATA

Future systems must prioritize scalability and efficiency, especially for large-scale and real-time applications. Research should explore lightweight deep learning architectures, approximate nearest neighbour search algorithms, and distributed indexing strategies to enhance performance while reducing computational overhead.

E. REDUCING COMPUTATIONAL COMPLEXITY AND RESOURCE DEMANDS

Optimizing algorithms for energy efficiency and low-resource environments is essential for real-time multimedia retrieval. Techniques like model compression, quantization, and edge computing deployment should be explored to make systems more adaptable to resource-constrained devices.

F. ADAPTIVE MODELS FOR DYNAMIC AND EVOLVING DATA

Retrieval systems must become more adaptive to dynamic datasets and evolving multimedia content. Future work should focus on online learning, incremental model updates, and self-supervised learning techniques to ensure systems remain effective in continuously changing data environments.

G. IMPROVING ROBUSTNESS TO NOISE AND DATA VARIABILITY

Enhancing the robustness of retrieval models against noise, domain shifts, and data heterogeneity is crucial. Research should explore adversarial training, domain adaptation, and noise-resistant feature extraction methods to improve system resilience and generalization.

H. INTEGRATING SECURITY AND PRIVACY-PRESERVING MECHANISMS

Given the increasing use of cloud and IoT-based retrieval systems, incorporating secure and privacy-preserving techniques is essential. Future research should focus on encrypted indexing, federated learning, and secure multi-party computation to safeguard user data while maintaining retrieval efficiency.

I. ESTABLISHING STANDARDIZATION AND BENCHMARKING PROTOCOLS

The absence of standardized benchmarks impedes consistent evaluation of retrieval systems. Developing universally accepted datasets, evaluation metrics, and benchmarking frameworks will enable fair comparisons and drive progress in multimedia retrieval research.

J. ADVANCING FEATURE REPRESENTATION AND FUSION TECHNIQUES

Future work should develop more comprehensive feature representations and fusion methods that capture complex relationships in multimedia data. Combining handcrafted features with deep learning representations can create more discriminative and robust models.

K. EFFICIENT HANDLING OF MULTILABEL AND COMPLEX ANNOTATIONS

Multimedia datasets often contain complex and multilabel annotations. Research should focus on designing models capable of efficiently handling multilabel classification and capturing intricate semantic relationships, possibly through hierarchical learning and semantic embedding techniques.

L. PROMOTING HUMAN-AI COLLABORATION IN ANNOTATION AND RETRIEVAL

Integrating human expertise with AI models can improve the accuracy and relevance of multimedia retrieval. Developing interactive systems that incorporate user feedback and human-in-the-loop learning strategies will enhance semantic annotation and retrieval effectiveness.

The future of multimedia retrieval is heading towards self-learning, privacy-preserving, and explainable AI models as per the trend forecast, as shown in Fig. 2. In 2023, we will see Zero-Shot Learning, where retrieval systems can identify unknown categories with minimal supervision, reducing dependency on labeled data. In 2024, Self-Supervised Learning will enhance retrieval accuracy by learning feature representations from unlabeled multimedia data. In 2025, Federated Retrieval will focus on private search, decentralized multimedia indexing without compromising user data. In 2026, Quantum Search will revolutionize similarity search by using quantum computing for superspeed, large-scale search. And in 2027, we will be focusing on Explainable AI (XAI) in Retrieval to ensure fairness, transparency, and bias mitigation in search results. This roadmap outlines the evolution of retrieval towards smarter, privacy-aware and ethically responsible AI-based systems. By addressing these future research directions, multimedia retrieval systems can become more intelligent, scalable, secure, and user-centric, ultimately advancing their applicability in diverse real-world scenarios.

1) RETRIEVAL-AUGMENTED GENERATION (RAG) AND PROMPT-DRIVEN MULTIMODAL SEARCH

While explainable AI and federated architectures offer valuable avenues, future research must move beyond these recurring themes to explore novel frameworks that unify retrieval with generative reasoning. One such direction is Retrieval-Augmented Generation (RAG) for multimodal inputs, which can bridge information gaps via dynamic querying fused with generative synthesis. Similarly, causal representation learning in retrieval systems can improve model robustness and debias cross-modal alignments. These paradigms demand an interdisciplinary approach combining vision-language modeling, human-computer interaction, and knowledge-grounded reasoning — an integration largely absent from existing literature.

An emerging frontier in multimodal retrieval is the integration of retrieval-augmented generation (RAG) frameworks, which combine classical retrieval methods with large-scale generative models. Rather than merely returning a ranked list of results, RAG-based systems retrieve relevant data (text, image, video, etc.) and use generative models—such as GPT-4V, BLIP-2, or Gemini—to synthesize responses, create multimodal summaries, or answer natural language queries with visual grounding. This paradigm enhances interaction quality and provides more contextualized and human-like outputs.

Complementing this shift is the rise of prompt engineering as a critical component in multimodal retrieval. Vision-language models like MiniGPT-4 and BLIP-2 demonstrate how carefully designed prompts can tailor retrieval across modalities. Prompt-tuned systems enable expressive querying, few-shot learning, and task-specific adaptations, even in domains with limited labeled data.

Future research directions should focus on:

- Designing modality-aware retrieval agents that combine learned memory with generative reasoning.
- Developing dynamic prompt-tuning strategies for domain-specific use cases such as clinical diagnostics, disaster response, or legal evidence retrieval.
- Addressing alignment and consistency challenges between retrieved content and generated responses, especially in multimodal output spaces.
- Exploring the use of RAG for real-time video summarization and cross-modal explanation generation.

This perspective repositions retrieval not as a terminal step but as a cognitive preprocessor for generative AI systems, marking a shift toward interactive, explainable, and adaptive multimedia intelligence.

2) ETHICAL CONSIDERATIONS IN MULTIMODAL RETRIEVAL

As multimodal retrieval systems become increasingly embedded in sensitive domains such as healthcare, law enforcement, education, and surveillance, ethical concerns surrounding their deployment must be foregrounded. One key issue is bias amplification in cross-modal alignment, where pretrained models (e.g., CLIP, BLIP-2) may inherit or even magnify societal biases from their training data—leading to skewed or discriminatory retrieval outcomes. For instance, image-text alignment models may exhibit racial, gender, or cultural stereotypes if not carefully audited.

Equally concerning are privacy vulnerabilities in decentralized and federated architectures. While federated learning frameworks are designed to ensure data locality and reduce leakage, recent studies reveal that reconstruction attacks and gradient inversion techniques can still compromise sensitive information. Moreover, retrieval systems that operate on biometric, medical, or behavioral modalities raise additional challenges around consent, data ownership, and traceability.

Future research must prioritize:

- Bias auditing protocols for multimodal datasets and learned embeddings,
- Differential privacy techniques integrated with federated retrieval models,
- Explainability-by-design, where models justify their retrieved results across modalities, and
- Ethical dataset curation that ensures inclusivity, transparency, and fairness.

Without these safeguards, the scalability of multimodal retrieval systems may come at the cost of societal trust, legal compliance, and human dignity.

3) BRIDGING RETRIEVAL WITH GENERATIVE AND CAUSAL AI

As retrieval systems move toward open-domain deployment, interdisciplinary integrations are becoming essential. One such frontier is the intersection of retrieval and generative AI, where prompt-based generation, retrieval-augmented generation (RAG), and text-to-image synthesis enable interactive and explanatory retrieval. Models like GPT-4V and BLIP-2

exemplify this blend, offering both retrieval and generative reasoning in a single multimodal framework.

Another important connection is with causal inference, especially for bias mitigation in multimodal sentiment or opinion-based retrieval. Recent approaches model causal pathways between visual and textual inputs (e.g., cause of sentiment) to eliminate spurious correlations. This is particularly relevant in applications such as social media monitoring, medical diagnostics, and policy surveillance, where understanding why an image-text pair aligns matters as much as the match itself.

To further illustrate the expanding boundaries of multimodal retrieval research, we highlight emerging interdisciplinary paradigms that fuse retrieval with symbolic reasoning, generative modeling, and causal inference. These approaches not only enhance retrieval capabilities but also address long-standing challenges such as explainability, fairness, and generalization. Table 18 summarizes representative synergies across these domains, showcasing their unique contributions to retrieval system design and deployment in real-world scenarios.

4) EMERGING ARCHITECTURAL PARADIGMS

While earlier systems emphasized accuracy and modality richness, newer paradigms prioritize deployment feasibility in real-world environments. Edge-AI optimizations—such as pruning, quantization, and lightweight transformer distillation (e.g., TinyCLIP [73])—are enabling real-time, on-device retrieval for mobile and IoT platforms. Simultaneously, real-time retrieval pipelines now utilize asynchronous indexing, streaming fusion modules, and vector compression frameworks (e.g., FAISS [74]) to meet low-latency demands in surveillance and AR/VR applications. Additionally, quantum-inspired approaches have emerged [75], leveraging probabilistic superposition and embeddings to explore richer semantic relationships in high-dimensional spaces. While still in nascent stages, these innovations represent a shift from static architectures toward dynamic, resource-aware, and potentially non-classical retrieval systems.

5) HYBRID RETRIEVAL-AUGMENTED SYSTEMS

A new class of hybrid retrieval-augmented systems is transforming how information is accessed and synthesized across modalities. These systems combine traditional indexing with large pretrained generative models (e.g., GPT-4V, BLIP-2), where external retrieval modules feed context into prompt-driven architectures. In the multimodal domain, this allows for queries like “describe scenes similar to this image” or “retrieve all videos where the object action leads to an outcome,” which require cross-modal memory fusion. Models such as REVEAL [76] and BLIP-RAG [77] exemplify this hybrid integration by combining retrieval precision with generative reasoning. However, such architectures introduce trade-offs in terms of latency, interpretability, and retrieval traceability, especially in open-ended

or instruction-based querying. Further research is needed to optimize retriever-generator alignment and manage semantic drift in real-time settings.

6) A UNIFIED ARCHITECTURE FOR FUTURE MULTIMEDIA RETRIEVAL

The proposed approach, as shown in Figure 3, unites fundamental principles such as self-learning as well as explainable AI, and privacy preservation into a single multimedia retrieval system. The fundamental module within the system applies state-of-the-art models, including CLIP, ViTs, and BLIP, through which it transforms multimodal inputs from images, videos, audio, and text into a unified semantic domain. The data moves through a sequence of operations, beginning from query processing until it reaches indexing, before retrieval, and ends with result presentation. The implementation optimizes steps specifically for scalability benefits with relevance criteria to achieve precise and fast retrieval within dynamic large database systems. The framework delivers customized outputs that serve healthcare applications, together with autonomous vehicles applications and e-commerce, as well as surveillance applications, because it demonstrates practical adaptability. The framework unites crucial components to resolve contemporary research problems related to adaptability, transparency, and secure management of multimedia information.

REFERENCES

- [1] M. Flickner, H. Sawhney, W. Niblack, J. Ashley, Q. Huang, B. Dom, M. Gorkani, J. Hafner, D. Lee, D. Petkovic, D. Steele, and P. Yanker, “Query by image and video content: The QBIC system,” *Computer*, vol. 28, no. 9, pp. 23–32, Sep. 1995, doi: [10.1109/2.410146](https://doi.org/10.1109/2.410146).
- [2] A. W. M. Smeulders, M. Worringer, S. Santini, A. Gupta, and R. Jain, “Content-based image retrieval at the end of the early years,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 12, pp. 1349–1380, Dec. 2000, doi: [10.1109/34.895972](https://doi.org/10.1109/34.895972).
- [3] R. Datta, D. Joshi, J. Li, and J. Z. Wang, “Image retrieval: Ideas, influences, and trends of the new age,” *ACM Comput. Surveys*, vol. 40, no. 2, pp. 1–60, Apr. 2008, doi: [10.1145/1348246.1348248](https://doi.org/10.1145/1348246.1348248).
- [4] M. S. Lew, N. Sebe, C. Djeraba, and R. Jain, “Content-based multimedia information retrieval: State of the art and challenges,” *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 2, no. 1, pp. 1–19, Feb. 2006, doi: [10.1145/1126004.1126005](https://doi.org/10.1145/1126004.1126005).
- [5] C. G. M. Snoek and M. Worringer, “Multimodal video indexing: A review of the state-of-the-art,” *Multimedia Tools Appl.*, vol. 25, no. 1, pp. 5–35, Jan. 2005, doi: [10.1023/B:MTAP.0000046380.27575.a5](https://doi.org/10.1023/B:MTAP.0000046380.27575.a5).
- [6] M. Naphade, J. R. Smith, J. Tesic, S.-F. Chang, W. Hsu, L. Kennedy, A. Hauptmann, and J. Curtis, “Large-scale concept ontology for multimedia,” *IEEE Multimedia Mag.*, vol. 13, no. 3, pp. 86–91, Jul. 2006.
- [7] A. Jaimes, and N. Sebe, “Multimodal human-computer interaction: A survey,” *Comput. Vis. Image Understand.*, vol. 108, no. 1–2, pp. 116–134, Jul. 2007, doi: [10.1016/J.CVIU.2006.10.019](https://doi.org/10.1016/J.CVIU.2006.10.019).
- [8] J. Liang, X. Ban, K. Yu, B. Qu, K. Qiao, C. Yue, K. Chen, and K. C. Tan, “A survey on evolutionary constrained multiobjective optimization,” *IEEE Trans. Evol. Comput.*, vol. 27, no. 2, pp. 201–221, Apr. 2023, doi: [10.1109/TEVC.2022.3155533](https://doi.org/10.1109/TEVC.2022.3155533).
- [9] R. Ashraf, M. Ahmed, U. Ahmad, M. A. Habib, S. Jabbar, and K. Naseer, “MDCBIR-MF: Multimedia data for content-based image retrieval by using multiple features,” *Multimedia Tools Appl.*, vol. 79, nos. 13–14, pp. 8553–8579, Apr. 2020, doi: [10.1007/s11042-018-5961-1](https://doi.org/10.1007/s11042-018-5961-1).
- [10] D. Srivastava, S. S. Singh, B. Rajitha, M. Verma, M. Kaur, and H.-N. Lee, “Content-based image retrieval: A survey on local and global features selection, extraction, representation, and evaluation parameters,” *IEEE Access*, vol. 11, pp. 95410–95431, 2023.

- [11] J. Pradhan, C. Bhaya, A. K. Pal, and A. Dhuriya, "Content-based image retrieval using DNA transcription and translation," *IEEE Trans. Nanobiosci.*, vol. 22, no. 1, pp. 128–142, Jan. 2023.
- [12] K. N. Sukhia, S. S. Ali, M. M. Riaz, A. Ghafoor, and B. Amin, "Content-based image retrieval using angles across scales," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5512005.
- [13] M. S. Lew, N. Sebe, C. Djeraba, and R. Jain, "Content-based multimedia information retrieval," *ACM Trans. Multimedia Comput., Commun. Appl.*, vol. 2, no. 1, pp. 1–19, Feb. 2006.
- [14] N. Rasiwasia, J. Costa Pereira, E. Coviello, G. Doyle, G. R. G. Lanckriet, R. Levy, and N. Vasconcelos, "A new approach to cross-modal multimedia retrieval," in *Proc. 18th ACM Int. Conf. Multimedia*, Oct. 2010, pp. 251–260.
- [15] M. Mahmood, W. J. Al-Kubaisy, and B. Al-Khateeb, "Using artificial neural network for multimedia information retrieval," *J. Southwest Jiaotong Univ.*, vol. 54, no. 3, pp. 1–16, Jun. 2019.
- [16] I. Bendib and M. R. Laouar, "A semantic indexing approach of multimedia documents content based partial transcription," in *Proc. 2nd Int. Conf. Natural Lang. Speech Process. (ICNLSP)*, Apr. 2018, pp. 1–6, doi: 10.1109/ICNLSP.2018.8374390.
- [17] J. Ahmad, K. Muhammad, J. Lloret, and S. W. Baik, "Efficient conversion of deep features to compact binary codes using Fourier decomposition for multimedia big data," *IEEE Trans. Ind. Informat.*, vol. 14, no. 7, pp. 3205–3215, Jul. 2018, doi: 10.1109/TII.2018.2800163.
- [18] M. Wang, W. Zhou, X. Yao, Q. Tian, and H. Li, "Towards codebook-free deep probabilistic quantization for image retrieval," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 1, pp. 626–640, Jan. 2024.
- [19] J.-M. Guo, A. W. H. Prayuda, H. Prasetyo, and S. Seshathiri, "Deep learning-based image retrieval with unsupervised double bit hashing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 7050–7065, Nov. 2023.
- [20] L. Jin, Z. Li, Y. Pan, and J. Tang, "Relational consistency induced self-supervised hashing for image retrieval," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 1482–1494, Jan. 2025.
- [21] K. Jiang, W. K. Wong, X. Fang, J. Li, J. Qin, and S. Xie, "Random online hashing for cross-modal retrieval," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 677–691, Jan. 2025.
- [22] Z. Zhang, F. Zhou, S. Qin, Q. Jia, and Z. Xu, "Privacy-preserving image retrieval and sharing in social multimedia applications," *IEEE Access*, vol. 8, pp. 66828–66838, 2020, doi: 10.1109/ACCESS.2020.2984916.
- [23] A. Du, L. Wang, S. Cheng, and N. Ao, "A privacy-protected image retrieval scheme for fast and secure image search," *Symmetry*, vol. 12, no. 2, p. 282, Feb. 2020.
- [24] Y. Ma, X. Chai, Z. Gan, and Y. Zhang, "Privacy-preserving TPE-based JPEG image retrieval in cloud-assisted Internet of Things," *IEEE Internet Things J.*, vol. 11, no. 3, pp. 4842–4856, Feb. 2024.
- [25] M. Liu, C. Yang, H. Li, and Y. Zhang, "An efficient attribute-based access control (ABAC) policy retrieval method based on attribute and value levels in multimedia networks," *Sensors*, vol. 20, no. 6, p. 1741, Mar. 2020, doi: 10.3390/s20061741.
- [26] S. Rateria and S. Singh, "Transparent, low resource, and context-aware information retrieval from a closed domain knowledge base," *IEEE Access*, vol. 12, pp. 44233–44243, 2024.
- [27] K. A. Hambarde and H. Proença, "Information retrieval: Recent advances and beyond," *IEEE Access*, vol. 11, pp. 76581–76604, 2023.
- [28] W. Chen, Y. Liu, W. Wang, E. M. Bakker, T. Georgiou, P. Fieguth, L. Liu, and M. S. Lew, "Deep learning for instance retrieval: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 7270–7292, Jun. 2023.
- [29] N. Gareshma, P. Vasuki, and S. Preethi, "Interactive audio indexing and speech recognition based navigation assist tool for tutoring videos," in *Proc. Int. Conf. Sustain. Comput. Data Commun. Syst. (ICSCDS)*, Erode, India, Apr. 2022, pp. 1678–1682, doi: 10.1109/icscds53736.2022.9760784.
- [30] S. Wagenpfeil, B. Vu, P. Mc Kevitt, and M. Hemmje, "Fast and effective retrieval for large multimedia collections," *Big Data Cognit. Comput.*, vol. 5, no. 3, p. 33, Jul. 2021, doi: 10.3390/bdcc5030033.
- [31] M. Kvet and J. Papan, "The complexity of the data retrieval process using the proposed index extension," *IEEE Access*, vol. 10, pp. 46187–46213, 2022, doi: 10.1109/ACCESS.2022.3170711.
- [32] Y. Zhu, C. Luo, T. Qian, L. Chen, Y. Gao, and B. Zheng, "DIMS: Distributed index for similarity search in metric spaces," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 1, pp. 210–225, Jan. 2025.
- [33] L. Han, M. E. Paoletti, X. Tao, Z. Wu, J. M. Haut, J. Plaza, and A. Plaza, "Central cohesion gradual hashing for remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 6002505, doi: 10.1109/LGRS.2023.3241849.
- [34] R. Imbriaco, C. Sebastian, E. Bondarev, and P. H. N. de With, "Toward multilabel image retrieval for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4703214.
- [35] Y. Zhang, X. Zheng, and X. Lu, "Remote sensing image retrieval by deep attention hashing with distance-adaptive ranking," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4301–4311, 2023.
- [36] M. Sun, "An intelligent retrieval method for audio and video content: Deep learning technology based on artificial intelligence," *IEEE Access*, vol. 12, pp. 123430–123446, 2024.
- [37] Q. Li, S. An, L. Li, W. Liu, and Y. Shao, "Multi-view diffusion process for spectral clustering and image retrieval," *IEEE Trans. Image Process.*, vol. 32, pp. 4610–4620, 2023.
- [38] Z. Liu, F. Chen, J. Xu, W. Pei, and G. Lu, "Image-text retrieval with cross-modal semantic importance consistency," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 5, pp. 2465–2476, May 2023.
- [39] S. Maharjan, S. Zhao, C. Zhong, and S. Jiang, "From LeanStore to LearnedStore: Using a learned index to improve database index search," in *Proc. Int. Conf. High Perform. Big Data Intell. Syst. (HDIS)*, Macau, China, Dec. 2023, pp. 162–169.
- [40] P. Steinert, S. Wagenpfeil, I. Frommholz, and M. L. Hemmje, "Towards the integration of metaverse and multimedia information retrieval," in *Proc. IEEE Int. Conf. Metrol. eXtended Reality, Artif. Intell. Neural Eng.*, Milano, Italy, Oct. 2023, pp. 581–586.
- [41] B. Aslam, A. Maqsoom, A. H. Cheema, F. Ullah, A. Alharbi, and M. Imran, "Water quality management using hybrid machine learning and data mining algorithms: An indexing approach," *IEEE Access*, vol. 10, pp. 119692–119705, 2022.
- [42] R. Fernandes, A. Pessoa, M. Salgado, A. De Paiva, I. Pacal, and A. Cunha, "Enhancing image annotation with object tracking and image retrieval: A systematic review," *IEEE Access*, vol. 12, pp. 79428–79444, 2024.
- [43] M. Arslan, M. Asad, A. H. Khan, S. Iqbal, M. N. Asghar, and A. A. Alaulamie, "Deep image synthesis, analysis and indexing using integrated CNN architectures," *IEEE Access*, vol. 13, pp. 834–851, 2025.
- [44] Y. Yang, H. Wang, J. Wang, K. Dong, and S. Ding, "Semantic-preserving surgical video retrieval with phase and behavior coordinated hashing," *IEEE Trans. Med. Imag.*, vol. 43, no. 2, pp. 807–819, Feb. 2024, doi: 10.1109/TMI.2023.3321382.
- [45] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
- [46] Y. Zhang, X. Zheng, and X. Lu, "Remote sensing image retrieval by deep attention hashing with distance-adaptive ranking," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4301–4311, 2023, doi: 10.1109/JSTARS.2023.3271303.
- [47] Y. Jia, X. Wu, H. Li, Q. Zhang, Y. Hu, S. Zhao, and W. Fan, "Uni-retrieval: A multi-style retrieval framework for STEM's education," 2025, *arXiv:2502.05863*.
- [48] G. Zhang, S. Li, S. Wei, S. Ge, N. Cai, and Y. Zhao, "Multimodal composition example mining for composed query image retrieval," *IEEE Trans. Image Process.*, vol. 33, pp. 1149–1161, 2024, doi: 10.1109/TIP.2024.3359062.
- [49] L. Xiao, R. Mao, S. Zhao, Q. Lin, Y. Jia, L. He, and E. Cambria, "Exploring cognitive and aesthetic causality for multimodal aspect-based sentiment analysis," *IEEE Trans. Affect. Comput.*, early access, Apr. 29, 2025, doi: 10.1109/TAFFC.2025.3565506.
- [50] S. Zerr, S. Siersdorfer, and J. Hare, "PicAlert!: A system for privacy-aware image classification and retrieval," in *Proc. 21st ACM Int. Conf. Inf. Knowl. Manage. (CIKM)*, New York, NY, USA: ACM, 2012, pp. 2710–2712, doi: 10.1145/2396761.2398735.
- [51] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.
- [52] S. M. Lundberg and S. I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–18.

- [53] M. Madheswaran and A. M. Murugaiyan, "A systematic video indexing approach using decision tree," *Int. J. Adv. Signal Image Sci.*, vol. 8, no. 2, pp. 18–25, Dec. 2022.
- [54] G. Amato, P. Bolettieri, F. Carrara, F. Falchi, C. Gennaro, N. Messina, L. Vadicamo, and C. Vairo, "VISIONE for newbies: An easier-to-use video retrieval system," in *Proc. 20th Int. Conf. Content-based Multimedia Indexing*, Sep. 2023, pp. 158–162.
- [55] F. Yang, N. A. Ismail, Y. Y. Pang, V. R. Kebande, A. Al-Dhaqm, and T. W. Koh, "A systematic literature review of deep learning approaches for sketch-based image retrieval: Datasets, metrics, and future directions," *IEEE Access*, vol. 12, pp. 14847–14869, 2024.
- [56] Q. Zou, L. Cao, Z. Zhang, L. Chen, and S. Wang, "Transductive zero-shot hashing for multilabel image retrieval," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1673–1687, Apr. 2022.
- [57] P. Mahalakshmi and N. S. Fatima, "Summarization of text and image captioning in information retrieval using deep learning techniques," *IEEE Access*, vol. 10, pp. 18289–18297, 2022.
- [58] M. H. Joseph and S. D. Ravana, "Reliable information retrieval systems performance evaluation: A review," *IEEE Access*, vol. 12, pp. 51740–51751, 2024.
- [59] S. Iqbal, A. N. Qureshi, M. Alhussein, I. A. Choudhry, K. Aurangzeb, and T. M. Khan, "Fusion of textual and visual information for medical image modality retrieval using deep learning-based feature engineering," *IEEE Access*, vol. 11, pp. 93238–93253, 2023.
- [60] R. Yelchuri, A. O. Khadidos, A. O. Khadidos, A. M. Alshareef, G. Swain, and J. K. Dash, "Deep semantic feature reduction for efficient remote sensing image retrieval," *IEEE Access*, vol. 11, pp. 112787–112803, 2023.
- [61] Z. Zhang, L. Wang, Y. Wang, L. Zhou, J. Zhang, and F. Chen, "Dataset-driven unsupervised object discovery for region-based instance image retrieval," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 247–263, Jan. 2023, doi: [10.1109/TPAMI.2022.3141433](https://doi.org/10.1109/TPAMI.2022.3141433).
- [62] F. Ye, W. Luo, M. Dong, D. Li, and W. Min, "Content-based remote sensing image retrieval based on fuzzy rules and a fuzzy distance," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8002505.
- [63] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," 2022, *arXiv:2204.14198*.
- [64] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny. (2023). *MiniGPT-4: Enhancing Vision-Language Understanding With GPT-4*. [Online]. Available: <https://github.com/Vision-CAIR/MiniGPT-4>
- [65] S. Huang, L. Dong, W. Wang, Y. Hao, S. Singhal, S. Ma, T. Lv, L. Cui, O. Khan Mohammed, B. Patra, Q. Liu, K. Aggarwal, Z. Chi, J. Bjorck, V. Chaudhary, S. Som, X. Song, and F. Wei, "Language is not all you need: Aligning perception with language models," 2023, *arXiv:2302.14045*.
- [66] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," 2023, *arXiv:2301.12597*.
- [67] T. Lin and R. Sablatnig, "Enhancing historical image retrieval with compositional cues," 2024, *arXiv:2403.14287*.
- [68] X. Li, M. Xu, J. Xu, T. Weise, L. Zou, F. Sun, and Z. Wu, "Image retrieval using a deep attention-based hash," *IEEE Access*, vol. 8, pp. 142229–142242, 2020, doi: [10.1109/ACCESS.2020.3011102](https://doi.org/10.1109/ACCESS.2020.3011102).
- [69] S. Jin, "Unsupervised semantic deep hashing," 2018, *arXiv:1803.06911*.
- [70] J. Zhang, Y. Peng, and M. Yuan, "SCH-GAN: Semi-supervised cross-modal hashing by generative adversarial network," *IEEE Trans. Cybern.*, vol. 50, no. 2, pp. 489–502, Feb. 2020, doi: [10.1109/TCYB.2018.2868826](https://doi.org/10.1109/TCYB.2018.2868826).
- [71] W. Ji, Y. Wei, Z. Zheng, H. Fei, and T.-S. Chua, "Deep multimodal learning for information retrieval," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 9739–9741.
- [72] F. Chen, P. Huang, X. Ge, J. Huang, and Z. Bao, "Multimodal sentiment analysis based on causal reasoning," 2024, *arXiv:2412.07292*.
- [73] K. Wu, H. Peng, Z. Zhou, B. Xiao, M. Liu, L. Yuan, H. Xuan, M. Valenzuela, X. S. Chen, X. Wang, H. Chao, and H. Hu, "TinyCLIP: CLIP distillation via affinity mimicking and weight inheritance," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 21913–21923.
- [74] F. Groh, L. Ruppert, P. Wieschollek, and H. P. A. Lensch, "GGNN: Graph-based GPU nearest neighbor search," *IEEE Trans. Big Data*, vol. 9, no. 1, pp. 267–279, Feb. 2023.
- [75] M. Goldsmith, H. Saarinen, G. García-Pérez, J. Malmi, M. A. C. Rossi, and S. Maniscalco, "Link prediction with continuous-time classical and quantum walks," *Entropy*, vol. 25, no. 5, p. 730, Apr. 2023, doi: [10.3390/e25050730](https://doi.org/10.3390/e25050730).

- [76] Z. Hu, A. Iscen, C. Sun, Z. Wang, K.-W. Chang, Y. Sun, C. Schmid, D. A. Ross, and A. Fathi, "Reveal: Retrieval-augmented visual-language pre-training with multi-source multimodal knowledge memory," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23369–23379.
- [77] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. K. Tler, M. Lewis, W.-T. Yih, T. R. Chel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, Red Hook, NY, USA, 9459, pp. 9459–9474.



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